

Práctica 3

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1. Suponga que $X \sim \mathcal{U}(0, 1)$. Determine la función de densidad de probabilidad de $Y = e^{-X^2}$.
2. Suponga que $X \sim \text{Exp}(1)$. Determine la función de densidad de probabilidad de $U = 2e^{-X}$.
3. Suponga que $X \sim \mathcal{U}(0, 1)$. Determine la función de densidad de probabilidad de $Y = \frac{1}{X}$.
4. Suponga que $X \sim \text{Exp}(1)$. Se define la función:

$$g(X) = \begin{cases} X & \text{si } x < 1 \\ \frac{1}{X} & \text{si } x > 1 \end{cases}.$$

Determine la función de densidad de probabilidad de $Y = g(X)$.

5. Sea X una variable aleatoria con función de densidad de probabilidad

$$f_X(x) = \begin{cases} 6x(1-x) & \text{si } 0 < x < 1 \\ 0 & \text{en otro caso} \end{cases}.$$

Determine la función de densidad de probabilidad de $Y = X^3$.

6. Sea X una variable aleatoria con función de densidad de probabilidad

$$f_X(x) = \begin{cases} e^{-x} & \text{si } x \geq 0 \\ 0 & \text{en otro caso} \end{cases}.$$

Determine la función de densidad de probabilidad de $Y = \sqrt{X}$.

7. Sea $X \sim \text{Exp}(\lambda)$. Determine la función de densidad de probabilidad de $Y = \sqrt{X}$.
8. Suponga que X es una variable aleatoria con función de densidad de probabilidad doble exponencial con parámetro $\lambda = 1$, es decir,

$$f_X(x) = \begin{cases} \frac{1}{2}e^{-|x|} & \text{si } -\infty < x < \infty \\ 0 & \text{en otro caso} \end{cases}.$$

Considere $Y = \mu + \sigma X$, para $\sigma > 0$. Determine la función de densidad de probabilidad de Y .

9. Suponga que $X \sim \mathcal{U}\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Determine la función de densidad de probabilidad de $Y = \tan(X)$.
10. La distribución de Cauchy es un ejemplo de una distribución definida, pero para la cual tanto el valor esperado como la variancia y la función generadora de momentos no están definidos. Esta distribución se usa para modelos de crecimiento y de eventos extremos en hidrología. Si X es una variable aleatoria con distribución de Cauchy con parámetros μ real (denominado de ubicación) y $\gamma > 0$ (denominado de escala), es decir, $X \sim \text{Cau}(\mu, \gamma)$, entonces su función de densidad está definida por:

$$f_X(x) = \frac{1}{\pi\gamma \left(1 + \left(\frac{x-\mu}{\gamma}\right)^2\right)}$$

y su función de distribución por:

$$F_X(x) = \frac{1}{\pi} \arctan\left(\frac{x-\mu}{\gamma}\right) + \frac{1}{2}$$

a) Si $X \sim \mathcal{U}(0, 1)$, encuentre la distribución de $Y = \tan(\pi(X - 1/2))$.

b) Si $X \sim \text{Cau}(0, 1)$, encuentre la distribución de $Y = rX + s$, con r y s constantes reales.

c) Si $X \sim \text{Cau}(\mu_0, \gamma_0)$ y $Y \sim \text{Cau}(\mu_1, \gamma_1)$ independientes, demuestre que $X + Y \sim \text{Cau}(\mu_0 + \mu_1, \gamma_0 + \gamma_1)$ y que $X - Y \sim \text{Cau}(\mu_0 - \mu_1, \gamma_0 + \gamma_1)$.

11. Sean X, Y dos variables aleatorias independientes distribuidas como $\Gamma(\alpha, 1)$ y $\Gamma(\beta, 1)$, respectivamente.

Considere las variables $U = \frac{X}{X+Y}$ y $V = X + Y$.

a) Determine la función de densidad de probabilidad conjunta de U y V .

b) Determine $f_U(u)$ y $f_V(v)$.

12. Sea $X \sim \mathcal{U}(0, 1)$ y Y con función de densidad de probabilidad

$$f_Y(y) = \begin{cases} 2y & \text{si } 0 < y < 1 \\ 0 & \text{en otro caso} \end{cases}$$

Suponga que son independientes.

a) Determine la función de densidad de probabilidad de $U = XY$.

b) Determine la función de densidad de probabilidad de $U = \frac{X}{Y}$.

13. Sean X, Y dos variables aleatorias con función de probabilidad conjunta

$$f_{X,Y}(x, y) = \begin{cases} c(x+y)e^{-x-y} & \text{si } x > 0, y > 0 \\ 0 & \text{en otro caso} \end{cases}$$

a) Determine el valor de la constante c .

b) Determine la función de densidad de probabilidad de $U = X + Y$.

14. Sean X, Y dos variables aleatorias con función de probabilidad conjunta

$$f_{X,Y}(x, y) = \begin{cases} cxy & \text{si } 0 < x < y < 1 \\ 0 & \text{en otro caso} \end{cases}$$

a) Determine el valor de la constante c .

b) Determine la función de densidad de probabilidad de $U = XY$.

15. Sean X, Y dos variables aleatorias con función de probabilidad conjunta

$$f_{X,Y}(x, y) = \begin{cases} 2e^{-x-2y} & \text{si } x > 0, y > 0 \\ 0 & \text{en otro caso} \end{cases}$$

a) Determine la función de densidad de probabilidad de $U = X + Y$.

b) Determine la función de densidad de probabilidad de $V = X - Y$.

16. Sean X, Y dos variables aleatorias con función de distribución conjunta

$$F_{X,Y}(x,y) = \begin{cases} x^3 y^2 & \text{si } x \geq 0, y \leq 1 \\ 0 & \text{en otro caso} \end{cases}.$$

a) Determine la función de densidad de probabilidad de $U = XY$.

b) Determine la función de densidad de probabilidad de $V = \frac{X}{Y}$.

17. Sean las variables aleatorias independientes $X \sim \mathcal{U}(0,1)$ y $Y \sim \text{Exp}(\lambda)$. Determine la función de densidad de probabilidad de $U = X + Y$.

18. Sean las variables aleatorias independientes $X \sim \text{Exp}(\lambda)$ y $Y \sim \text{Exp}(\mu)$. Determine la función de densidad de probabilidad de cada una de las variables aleatorias $Z = X + Y$ y $W = X - Y$.

19. Sean X, Y dos variables aleatorias con función de densidad conjunta

$$f_{X,Y}(x,y) = \begin{cases} 6e^{-3x-2y} & \text{si } x \geq 0, y \geq 0 \\ 0 & \text{en otro caso} \end{cases}.$$

Determine la función de densidad de probabilidad de $U = X + Y$.

20. Sean X, Y dos variables aleatorias con función de densidad conjunta

$$f_{X,Y}(x,y) = \begin{cases} 5xe^{-xy} & \text{si } 0.2 \leq x \leq 0.4, y \geq 0 \\ 0 & \text{en otro caso} \end{cases}.$$

Determine la función de densidad de probabilidad de $U = XY$.

21. Sean X, Y dos variables aleatorias con función de densidad conjunta

$$f_{X,Y}(x,y) = \begin{cases} e^{-x-y} & \text{si } x \geq 0, y \geq 0 \\ 0 & \text{en otro caso} \end{cases}.$$

Si $U = \frac{X}{Y}$ y $V = X + Y$, determine la función de densidad de probabilidad conjunta de U y V .

22. Sean las variables aleatorias independientes $X \sim \mathcal{N}(0,1)$, $Y \sim \Gamma(\frac{1}{2}, 2)$. Si $U = \frac{X}{\sqrt{Y}}$ y $V = Y$, determine $f_{U,V}(u,v)$ y $f_U(u)$.

23. Sean las variables aleatorias X, Y independientes e idénticamente distribuidas como $\mathcal{N}(0,1)$. Si $U = 2X - Y$ y $V = -X + Y$, determine $f_{Z,W}(z,w)$.

24. Si $Y \sim \Gamma(\alpha, 1)$, muestre que $W = \lambda Y \sim \Gamma(\alpha, \lambda)$.

25. Si $Y \sim \mathcal{U}(0,1)$, muestre que $W = -\ln(Y) \sim \text{Exp}(1)$.

26. Si $Y \sim \text{Pareto}(\alpha, \beta)$, es decir, Y tiene una función de densidad de probabilidad

$$f_Y(y) = \frac{\alpha \beta^\alpha}{y^{\alpha+1}}, \quad y \geq \beta > 0, \alpha > 0$$

Muestre que $W = \ln\left(\frac{Y}{\beta}\right) \sim \text{Exp}(\alpha)$.

27. Si $Y \sim \text{Exp}(\lambda)$. ¿Cómo se distribuye $W = 2\lambda Y$?

28. Sean $X_1, \dots, X_n$, n variables aleatorias independientes e idénticamente distribuidas, tal que $X_i \sim \text{Exp}(\lambda)$.

¿Cómo se distribuye la variable aleatoria $U = \frac{2}{\lambda} \sum_{i=1}^n X_i$?

29. Sean $X_1, \dots, X_n$, n variables aleatorias independientes e idénticamente distribuidas con función de densidad de probabilidad de Rayleigh

$$f_X(x) = \frac{x}{\theta^2} e^{-\frac{x^2}{2\theta^2}}, \quad \theta > 0, x > 0$$

¿Cómo se distribuye la variable aleatoria $U = \frac{1}{\theta^2} \sum_{i=1}^n X_i^2$?

30. Sean $X_1, \dots, X_n$, n variables aleatorias independientes e idénticamente distribuidas como $\text{Pareto}(\alpha, \beta)$.

$$f_Y(y) = \frac{\alpha \beta^\alpha}{y^{\alpha+1}}, \quad y \geq \beta > 0, \alpha > 0$$

a) Muestre que el mínimo muestral $Y_{(1)} \sim \text{Pareto}(n\alpha, \beta)$.

b) ¿Cómo se distribuye la variable aleatoria $U = \frac{Y_{(1)}}{\beta}$?

Funciones continuas

$$\frac{-1}{\infty} \circ \frac{1}{\infty} = 0$$

Uniforme $X \sim U(a, b)$

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{si } a \leq x \leq b \\ 0 & \text{en o.c.} \end{cases}$$

$$\frac{1}{0} = \infty$$

$$F_X(x) = \begin{cases} 0 & \text{si } x < a \\ \frac{x-a}{b-a} & \text{si } a \leq x \leq b \\ 1 & \text{si } x > b \end{cases}$$

$$m_X(t) = \begin{cases} \frac{e^{tb} - e^{ta}}{t(b-a)} \end{cases}$$

Normal $X \sim N(\mu, \sigma)$

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$m_X(t) = e^{t\mu + \frac{t^2\sigma^2}{2}}$$

Normal Estándar $X \sim Z(0,1)$.

$$f_X(x) = \begin{cases} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x^2}{2}} \\ 0 \text{ en o. c.} \end{cases}$$

$$m_X(t) = e^{-\frac{t^2}{2}}$$

Gamma $X \sim \Gamma(\alpha, \beta)$

$$f_X(x) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} \cdot x^{\alpha-1} \cdot e^{-x/\beta} & \text{si } x \geq 0 \\ 0 \text{ en o. c.} \end{cases}$$

$$\Gamma(1/2) = \sqrt{\pi}$$

$$\Gamma(\alpha) = (\alpha-1)! \quad \text{si } \alpha \in \mathbb{N} \quad m_X(t) = (1 - \beta t)^{-\alpha}$$

$$E(X) = \alpha\beta \quad \text{Var}(X) = \alpha\beta^2$$

Exponencial $X \sim \text{Exp}(\frac{1}{\beta})$

$$f_X(x) = \begin{cases} \frac{1}{\beta} \cdot e^{-x/\beta} & \text{si } x \geq 0 \\ 0 \text{ en o. c.} \end{cases}$$

$$F_X(x) = 1 - e^{-x/\beta}$$

$$E(X) = \beta \quad \text{Var}(X) = \beta^2$$

$$m_X(t) = (1 - \beta t)^{-1}$$

Beta $X \sim \text{Beta}(\alpha, \beta)$

$$f_X(x) = \begin{cases} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot x^{\alpha-1} (1-x)^{\beta-1} & \text{si } 0 \leq x \leq 1 \\ 0 & \text{en o. c.} \end{cases}$$

$$\frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} = \frac{(\alpha + \beta - 1)!}{(\alpha - 1)! (\beta - 1)!}$$

$$E(X) = \frac{\alpha}{\alpha + \beta} \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}$$

$m_X(t) =$ se omite.

Chi-cuadrado $X \sim \chi^2(V)$.

$$f_X(x) = \begin{cases} \frac{1}{2^{V/2} \Gamma(V/2)} \cdot x^{V/2-1} \cdot e^{-x/2} & \text{si } x \geq 0 \\ 0 & \text{en o. c.} \end{cases}$$

$V = GL$.

$$m_X(t) = (1 - 2t)^{-V/2}$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$Z = \frac{Y - \mu}{\sigma}, \quad Y \sim N(\mu, \sigma^2)$$

$$m_w(t) = e^{\frac{t^2}{2} \cdot \sigma^2} \quad \text{si } w = Y - \mu.$$

Distribución de Cauchy:

$$X \sim \text{Can}(\mu, r).$$

$$f_X(x) = \frac{1}{\pi r \left(1 + \left(\frac{x - \mu}{r} \right)^2 \right)}$$

$$X \sim \text{Can}(0, 1)$$

$$f_X(x) = \frac{1}{\pi (1 + x^2)}$$

$$X \sim \frac{1}{n} = \bar{X}$$

Transformation simple

1. $X \sim U(0,1)$. $Y = e^{-X^2}$. Determine $f_Y(y)$.

$$f_X(x) = \begin{cases} 1 & \text{si } 0 < x < 1 \\ 0 & \text{en o. c.} \end{cases}$$

Solucion:

$$Y = e^{-X^2}$$

$$\ln y = -X^2$$

$$\sqrt{-\ln y} = \sqrt{X^2}$$

$$\underbrace{\sqrt{-\ln y}}_{h^{-1}(y)} = X$$

$$\begin{aligned} \frac{dh^{-1}}{dy} &= -\ln y^{\frac{1}{2}} \\ &= \frac{1}{2} \cdot -\ln y^{-\frac{1}{2}} - \frac{1}{y} \\ &= -\frac{1}{2y\sqrt{-\ln y}} \end{aligned}$$

$$\left| \frac{dh^{-1}}{dy} \right| = \frac{1}{2y\sqrt{-\ln y}}$$

$$f_Y(y) = f_X(h^{-1}(y)) \cdot \left| \frac{dh^{-1}}{dy} \right|$$

$$= 1 \cdot \frac{1}{2y\sqrt{-\ln y}}$$

✓ Integra
a 1.

$$= \begin{cases} \frac{1}{2y\sqrt{-\ln y}} & \frac{1}{e} < y < 1 \\ 0 & \text{en o. c.} \end{cases}$$

Límites

$$Y = e^{-X^2}$$

$$\text{si } X=0 \Rightarrow Y=1$$

$$\text{si } X=1 \Rightarrow Y = e^{-1} = \frac{1}{e}$$

$$\therefore \frac{1}{e} < y < 1$$

Distribucion

1. $X \sim U(0,1)$. $Y = e^{-X^2}$. Determine $f_Y(y)$.

$$f_X(x) = \begin{cases} 1 & \text{si } 0 < x < 1 \\ 0 & \text{en o.c.} \end{cases}$$

Solucion

$$Y = e^{-X^2}$$

$$\text{si } x=0 \Rightarrow y=1$$

$$\text{si } x=1 \Rightarrow y = e^{-1} = \frac{1}{e}.$$

$$\therefore \frac{1}{e} < y < 1.$$

$$\Rightarrow \text{si } y < \frac{1}{e}$$

$$F_Y(y) = 0$$

$$\Rightarrow \text{si } \frac{1}{e} < y < 1$$

$$F_Y(y) = P(Y \leq y) = P(e^{-X^2} \leq y)$$

$$= P(-X^2 \leq \ln y)$$

$$= P(\sqrt{X^2} \geq -\ln y)$$

$$= P(X \geq \sqrt{-\ln y})$$

$$= 1 - P(X < \sqrt{-\ln y})$$

$$= 1 - \int_0^{\sqrt{-\ln y}} 1 \, dx.$$

$$= 1 - \int_0^{\sqrt{-\ln y}} 1 \, dx$$

$$= 1 - \sqrt{-\ln y}$$

$\Rightarrow$ si $y > 1$

$$F_Y(y) = 1$$

$$\therefore F_Y(y) = \begin{cases} 0 & \text{si } y < \frac{1}{e} \\ 1 - \sqrt{-\ln y} & \text{si } \frac{1}{e} < y < 1 \\ 1 & \text{si } y > 1 \end{cases}$$

$$f_Y(y) = \frac{dF_Y(y)}{dy} = 0 - \frac{1}{2}(-\ln y)^{\frac{1}{2}-1} \cdot -\frac{1}{y}$$

✓ Integra
a 1.

$$= \begin{cases} \frac{1}{2y\sqrt{-\ln y}} & \text{si } \frac{1}{e} < y < 1 \\ 0 & \text{en o. c} \end{cases}$$

Transformation
simple

2. $X \sim \text{Exp}(1)$. $U = 2e^{-X}$. Determine $f_U(u)$.

$$f_X(x) = \begin{cases} e^{-x} & \text{si } x \geq 0 \\ 0 & \text{en o.c.} \end{cases}$$

Solución

$$U = 2e^{-X}$$

$$\text{si } X=0 \Rightarrow U=2$$

$$\text{si } X=\infty \Rightarrow U=0$$

$$\left. \begin{array}{l} \text{si } X=0 \Rightarrow U=2 \\ \text{si } X=\infty \Rightarrow U=0 \end{array} \right\} 0 < U < 2$$

$$U = 2e^{-X}$$

$$\frac{U}{2} = e^{-X}$$

$$\ln\left(\frac{U}{2}\right) = -X$$

$$\underbrace{-\ln\left(\frac{U}{2}\right)}_{h^{-1}(U)} = X$$

$$\begin{aligned} \frac{dh^{-1}}{du} &= -[\ln(u) - \ln(2)] \\ &= -\left[\frac{1}{u} - 0\right] \\ &= -\frac{1}{u} \end{aligned}$$

$$\left| \frac{dh^{-1}}{du} \right| = \frac{1}{u}$$

$$f_U(u) = f_X(h^{-1}(u)) \cdot \left| \frac{dh^{-1}}{du} \right|$$

$$= e^{\ln\left(\frac{u}{2}\right)} \cdot \frac{1}{u}$$

$$\checkmark \text{ Integra } = \frac{u}{2} \cdot \frac{1}{u} = \frac{1}{2}$$

$$\text{a } 1. = \begin{cases} \frac{1}{2} & \text{si } 0 < u < 2, \\ 0 & \text{en o.c.} \end{cases}$$

Distribucion

2. $X \sim \text{Exp}(1)$. $U = 2e^{-X}$. Determine $f_U(u)$.

$$f_X(x) = \begin{cases} e^{-x} & \text{si } x \geq 0 \\ 0 & \text{en o. c.} \end{cases}$$

Solucion

$$U = 2e^{-X}$$

$$\text{si } X = 0 \Rightarrow U = 2$$

$$\text{si } X = \infty \Rightarrow U = 0$$

$$\left. \begin{array}{l} \text{si } X = 0 \Rightarrow U = 2 \\ \text{si } X = \infty \Rightarrow U = 0 \end{array} \right\} 0 < U < 2$$

$\Rightarrow$ si $U < 0$.

$$F_U(u) = 0.$$

$\Rightarrow$ si $0 < U < 2$.

$$F_U(u) = P(U \leq u) = P(2e^{-X} \leq u)$$

$$= P(e^{-X} \leq \frac{u}{2})$$

$$= P(-X \leq \ln(\frac{u}{2}))$$

$$= P(X \geq -\ln(\frac{u}{2}))$$

$$= 1 - P(X \leq -\ln(\frac{u}{2}))$$

$$= 1 - \int_0^{-\ln(\frac{u}{2})} e^{-x} dx$$

$$= 1 - \left(-e^{-x} \right) \Big|_0^{-\ln(\frac{u}{2})}$$

$$= 1 - \left(-e^{\ln(\frac{u}{2})} + 1 \right)$$

$$= 1 - (-e^{\ln(\frac{u}{2})} + 1)$$

$$= 1 - \left(1 - \frac{u}{2} + 1\right) = 1 + \frac{u}{2} - 1 = \boxed{\frac{u}{2}}$$

⇒ si $u > 2$

$$Fu(u) = 1$$

$$\therefore Fu(u) = \begin{cases} 0 & \text{si } u < 0 \\ \frac{u}{2} & \text{si } 0 < u < 2 \\ 1 & \text{si } u > 2 \end{cases}$$

$$fu(u) = \frac{dFu(u)}{du} = \frac{1 \cdot 2 - u \cdot 0}{4} = \frac{2}{4} = \boxed{\frac{1}{2}}$$

$$fu(u) = \begin{cases} \frac{1}{2} & \text{si } 0 < u < 2 \\ 0 & \text{en o. c.} \end{cases} \quad \text{r Integra a 1.}$$

Transformación simple

3. $X \sim U(0,1)$. $Y = \frac{1}{X}$. Determine $f_Y(y)$.

$$f_X(x) = \begin{cases} 1 & \text{si } 0 < x < 1 \\ 0 & \text{en o.c.} \end{cases}$$

Solución

$$Y = \frac{1}{X}$$

$$\left. \begin{array}{l} \text{si } X=0 \Rightarrow Y = \frac{1}{0} = \infty \\ \text{si } X=1 \Rightarrow Y = \frac{1}{1} = 1 \end{array} \right\} 1 < Y < \infty$$

$$Y = \frac{1}{X}$$

$$XY = 1$$

$$X = \frac{1}{Y}$$

$$h^{-1}(y)$$

$$\frac{dh^{-1}}{dy} = \frac{0 \cdot y - 1 \cdot 1}{Y^2}$$

$$= \frac{-1}{Y^2}$$

$$\left| \frac{dh^{-1}}{dy} \right| = \frac{1}{Y^2}$$

$$f_Y(y) = f_X(h^{-1}(y)) \cdot \left| \frac{dh^{-1}}{dy} \right|$$

$$= \begin{cases} \frac{1}{Y^2} & \text{si } Y \geq 1 \\ 0 & \text{en o.c.} \end{cases} \quad \checkmark \text{ Integra a 1.}$$

Distribucion

3. $X \sim U(0, 1)$. $Y = \frac{1}{X}$. Determine $f_Y(y)$.

$$f_X(x) = \begin{cases} 1 & \text{si } 0 < x < 1 \\ 0 & \text{en o.c.} \end{cases}$$

Solucion

$$Y = \frac{1}{X}$$

$$\left. \begin{array}{l} \text{si } x=0 \Rightarrow y = \frac{1}{0} = \infty \\ \text{si } x=1 \Rightarrow y = \frac{1}{1} = 1 \end{array} \right\} y \geq 1$$

$\Rightarrow$ si $y < 1$.

$$F_Y(y) = 0.$$

$\Rightarrow$ si $y \geq 1$

$$F_Y(y) = P(Y \leq y) = P\left(\frac{1}{X} \leq y\right)$$

$$= P(1 \leq XY)$$

$$= P\left(\frac{1}{y} \leq X\right)$$

$$= \int_{\frac{1}{y}}^1 1 \, dx$$

$$= x \Big|_{\frac{1}{y}}^1$$

$$= 1 - \frac{1}{y}$$

$$\therefore F_Y(y) = \begin{cases} 0 & \text{si } y < 1 \\ 1 - \frac{1}{y} & \text{si } y \geq 1 \end{cases}$$

$$f_Y(y) = \frac{d F_Y(y)}{dy} = 0 - \frac{0 \cdot y - 1 \cdot 1}{y^2} = 0 + \frac{1}{y^2} = \frac{1}{y^2}$$

$$f_Y(y) = \begin{cases} \frac{1}{y^2} & \text{si } y \geq 1 \\ 0 & \text{en o.c.} \end{cases} \quad \checkmark \text{ Integra a 1.}$$

4. $X \sim \text{Exp}(1)$. $f_X(X) = \begin{cases} e^{-x} & \text{si } x \geq 0 \\ 0 & \text{en o.c.} \end{cases}$

$$g(X) = \begin{cases} X & \text{si } X < 1 \\ \frac{1}{X} & \text{si } X > 1 \end{cases}$$

$Y = g(X)$. Determine $f_Y(Y)$.

Si $Y = X$, $0 < X < 1$.

$$\left. \begin{array}{l} \text{si } X=0 \Rightarrow Y=0. \\ \text{si } X=1 \Rightarrow Y=1 \end{array} \right\} 0 < Y < 1.$$

$\Rightarrow$ Si $Y < 0$.

$F_Y(Y) = 0$.

$\Rightarrow$ Si $0 < Y < 1$.

$F_Y(Y) = P(Y \leq y) = P(X \leq y)$

$$= \int_0^y e^{-x} dx.$$

$$= (-e^{-x} \big|_0^y)$$

$$\boxed{= -e^{-y} + 1.}$$

$$\text{Si } Y = \frac{1}{X}, \quad 1 < X < \infty$$

$$\left. \begin{array}{l} \text{Si } X=1 \Rightarrow Y = \frac{1}{1} = 1 \\ \text{Si } X=\infty \Rightarrow Y = \frac{1}{\infty} = 0 \end{array} \right\} 0 < Y < 1$$

$$F_Y(y) = P(Y \leq y) = P\left(\frac{1}{X} \leq y\right)$$

$$= P\left(\frac{1}{y} \leq X\right)$$

$$= \int_{\frac{1}{y}}^{\infty} e^{-x} dx$$

$$= \left(-e^{-x} \mid \frac{1}{y} \right)$$

$$= \left(0 + e^{-\frac{1}{y}} \right)$$

$$\boxed{= e^{-\frac{1}{y}}}$$

$$\therefore F_Y(y) = \begin{cases} 0 & \text{si } y < 0 \\ -e^{-y} + 1 + e^{-\frac{1}{y}} & \text{si } 0 < y < 1 \\ 1 & \text{si } y > 1 \end{cases}$$

$$\begin{aligned}
 f_Y(y) &= \frac{d F_Y(y)}{dy} = e^{-y} + 0 + e^{-\frac{1}{y}} \cdot \left(-\frac{1}{y}\right)' \\
 &= e^{-y} + e^{-\frac{1}{y}} \cdot \left(\frac{0 \cdot y + 1 \cdot 1}{y^2}\right) \\
 &= e^{-y} + e^{-\frac{1}{y}} \cdot \frac{1}{y^2}
 \end{aligned}$$

$$\sqrt{\text{Integral}} = \begin{cases} e^{-y} + \frac{e^{-\frac{1}{y}}}{y^2} & \text{si } 0 < y < 1 \\ 0 & \text{en o. c.} \end{cases}$$

Transformación
simple

$$5. f_X(x) = \begin{cases} 6x(1-x) & \text{si } 0 < x < 1 \\ 0 & \text{en o.c.} \end{cases}$$

$$Y = X^3. \text{ Determinar } f_Y(y).$$

$$Y = X^3.$$

$$\left. \begin{array}{l} \text{si } x=0 \Rightarrow Y=0^3=0 \\ \text{si } x=1 \Rightarrow Y=1^3=1 \end{array} \right\} 0 < y < 1$$

Solución:

$$\frac{dh^{-1}}{dy} = Y^{\frac{1}{3}} = \frac{1}{3} Y^{-\frac{2}{3}} = \frac{1}{3\sqrt[3]{y^2}}$$

$$Y = X^3$$

$$\underbrace{\sqrt[3]{Y}}_{h^{-1}(y)} = X$$

$$\left| \frac{dh^{-1}}{dy} \right| = \frac{1}{3\sqrt[3]{y^2}}$$

$$f_Y(y) = f_X(h^{-1}(y)) \cdot \left| \frac{dh^{-1}}{dy} \right|$$

$$= 6\sqrt[3]{y}(1-\sqrt[3]{y}) \cdot \frac{1}{3\sqrt[3]{y^2}}$$

✓ Integra a 1.

$$= \frac{2\sqrt[3]{y}(1-\sqrt[3]{y})}{\sqrt[3]{y^2}} = \begin{cases} \frac{2(1-\sqrt[3]{y})}{\sqrt[3]{y}} & \text{si } 0 < y < 1 \\ 0 & \text{en o.c.} \end{cases}$$

Distribucion

$$5. f_X(x) = \begin{cases} 6x(1-x) & \text{si } 0 < x < 1 \\ 0 & \text{en o.c.} \end{cases}$$

$Y = X^3$. Determine $f_Y(y)$.

$$Y = X^3$$

$$\left. \begin{array}{l} \text{si } x=0 \Rightarrow Y=0^3=0 \\ \text{si } x=1 \Rightarrow Y=1^3=1 \end{array} \right\} 0 < y < 1$$

Solucion

$$\Rightarrow \text{si } y < 0$$

$$F_Y(y) = 0.$$

$$\Rightarrow \text{si } 0 < y < 1$$

$$F_Y(y) = P(Y \leq y) = P(X^3 \leq y)$$

$$= P(X \leq \sqrt[3]{y})$$

$$= \int_0^{\sqrt[3]{y}} 6x - 6x^2 dx$$

$$= 6 \cdot \frac{x^2}{2} - 6 \cdot \frac{x^3}{3} \Big|_0^{\sqrt[3]{y}}$$

$$= 3(\sqrt[3]{y})^2 - 2 \cdot (\sqrt[3]{y})^3$$

$$\boxed{= 3(\sqrt[3]{y})^2 - 2y}$$

$$\Rightarrow \text{si } y > 1$$

$$F_Y(y) = 1.$$

$$\therefore F_Y(y) = \begin{cases} 0 & \text{si } y < 0 \\ 3(\sqrt[3]{y})^2 - 2y & \text{si } 0 < y < 1 \\ 1 & \text{si } y > 1. \end{cases}$$

$$f_Y(y) = \frac{dF_Y(y)}{dy} = 3 \cdot y^{\frac{2}{3}} - 2y$$

$$= \frac{6}{3} \cdot y^{-\frac{1}{3}} - 2$$

$$= \frac{2}{\sqrt[3]{y}} - \frac{2}{1}$$

$$= \frac{12 - 2\sqrt[3]{y}}{\sqrt[3]{y}}$$

$$\text{Integral a 1.} = \begin{cases} \frac{2(1 - \sqrt[3]{y})}{\sqrt[3]{y}} & \text{si } 0 < y < 1 \\ 0 & \text{en o. c.} \end{cases}$$

Transformación
simple

$$6. f_X(x) = \begin{cases} e^{-x} & \text{si } x \geq 0 \\ 0 & \text{en o.c.} \end{cases}$$

$Y = \sqrt{X}$. Determine $f_Y(y)$.

$$Y = \sqrt{X}$$

$$\text{si } x=0 \Rightarrow y = \sqrt{0} = 0.$$

$$\text{si } x=\infty \Rightarrow y = \sqrt{\infty} = \infty$$

$$\left. \begin{array}{l} \text{si } x=0 \Rightarrow y = \sqrt{0} = 0. \\ \text{si } x=\infty \Rightarrow y = \sqrt{\infty} = \infty \end{array} \right\} y \geq 0.$$

Solución

$$Y = \sqrt{X}$$

$$Y^2 = X$$
$$\underbrace{\hspace{1cm}}_{h^{-1}(y)}$$

$$\frac{dh^{-1}}{dy} = 2y$$

$$\left| \frac{dh^{-1}}{dy} \right| = 2y$$

$$f_Y(y) = f_X(h^{-1}(y)) \cdot \left| \frac{dh^{-1}}{dy} \right|$$

$$= e^{-y^2} \cdot 2y$$

$$= \begin{cases} 2y \cdot e^{-y^2} & \text{si } y \geq 0 \\ 0 & \text{en o.c.} \end{cases} \quad \checkmark \text{Integra a 1.}$$

Distribucion

$$6. f_X(x) = \begin{cases} e^{-x} & \text{si } x \geq 0 \\ 0 & \text{en o.c.} \end{cases}$$

$Y = \sqrt{X}$. Determine $f_Y(y)$.

$$Y = \sqrt{X}.$$

$$\left. \begin{array}{l} \text{si } x=0 \Rightarrow Y = \sqrt{0} = 0 \\ \text{si } x=\infty \Rightarrow Y = \sqrt{\infty} = \infty \end{array} \right\} Y \geq 0.$$

$$\text{si } Y < 0$$

$$\Rightarrow F_Y(Y) = 0.$$

$$\text{si } Y \geq 0.$$

$$\Rightarrow F_Y(Y) = P(Y \leq y) = P(\sqrt{X} \leq y).$$

$$= P(X \leq y^2)$$

$$= \int_0^{y^2} e^{-x} dx$$

$$= -e^{-x} \Big|_0^{y^2}.$$

$$= -e^{-y^2} + 1$$

$$\boxed{= 1 - e^{-y^2}}$$

$$\therefore F_Y(y) = \begin{cases} 0 & \text{si } y < 0. \\ 1 - e^{-y^2} & \text{si } y \geq 0. \\ 0 & \text{en o. c.} \end{cases}$$

$$f_Y(y) = \frac{dF_Y(y)}{dy} = 0 - -2y \cdot e^{-y^2} \\ = \begin{cases} 2y e^{-y^2} & \text{si } y \geq 0 \\ 0 & \text{en o. c.} \end{cases}$$

✓ Integra
a 1.

Transformación sencilla

7. $X \sim \text{Exp}(\lambda)$. $Y = \sqrt{X}$. Determine $f_Y(y)$.

$$f_X(x) = \begin{cases} \frac{1}{\lambda} \cdot e^{-x/\lambda} & \text{si } x \geq 0 \\ 0 & \text{en o.c.} \end{cases}$$

$$Y = \sqrt{X}$$

$$\left. \begin{array}{l} \text{si } x = 0 \Rightarrow Y = \sqrt{0} = 0 \\ \text{si } x = \infty \Rightarrow Y = \sqrt{\infty} = \infty \end{array} \right\} Y \geq 0$$

Solución

$$Y = \sqrt{X}$$

$$\underbrace{Y^2 = X}_{h^{-1}(Y)}$$

$$\frac{dh^{-1}}{dy} = 2y$$

$$\left| \frac{dh^{-1}}{dy} \right| = 2y$$

$$f_Y(y) = f_X(h^{-1}(y)) \cdot \left| \frac{dh^{-1}}{dy} \right|$$

$$= \frac{1}{\lambda} \cdot e^{-\frac{y^2}{\lambda}} \cdot 2y$$

$$= \begin{cases} \frac{2y}{\lambda} \cdot e^{-\frac{y^2}{\lambda}} & \text{si } y \geq 0 \\ 0 & \text{en o.c.} \end{cases}$$

✓ Integra a 1.

7. $X \sim \text{Exp}(\lambda)$ $Y = \sqrt{X}$.

Determine la $f_Y(y)$.

$$f_X(x) = \begin{cases} \frac{1}{\lambda} \cdot e^{-x/\lambda} & \text{si } x \geq 0 \\ 0 & \text{en o. c.} \end{cases}$$

$$Y = \sqrt{X}$$

$$\left. \begin{array}{l} \text{si } x=0 \Rightarrow Y = \sqrt{0} = 0 \\ \text{si } x=\infty \Rightarrow Y = \sqrt{\infty} = \infty \end{array} \right\} Y \geq 0$$

si $Y \leq 0$

$$\Rightarrow F_Y(Y) = 0.$$

si $Y \geq 0$.

$$\Rightarrow F_Y(Y) = P(Y \leq y) = P(\sqrt{X} \leq y)$$

$$= P(X \leq y^2)$$

$$= \int_0^{y^2} \frac{1}{\lambda} \cdot e^{-x/\lambda} dx.$$

$$= \frac{1}{\lambda} \left(\frac{e^{-x/\lambda}}{-\frac{1}{\lambda}} \bigg|_0^{y^2} \right)$$

$$= \frac{1}{\lambda} \left(-\lambda e^{-x/\lambda} \bigg|_0^{y^2} \right)$$

$$= \frac{1}{\lambda} \left(-\lambda e^{-\frac{y^2}{\lambda}} + \lambda \right)$$

$$\boxed{= -e^{-\frac{y^2}{\lambda}} + 1}$$

$$\therefore F_Y(y) = \begin{cases} 0 & \text{si } y < 0 \\ 1 - e^{-\frac{y^2}{\lambda}} & \text{si } y \geq 0 \end{cases}$$

$$\begin{aligned} f_Y(y) &= \frac{dF_Y(y)}{dy} = 0 - e^{-\frac{y^2}{\lambda}} \cdot \left(\frac{-2y \cdot \lambda + y^2}{\lambda^2} \right) \\ &= -e^{-\frac{y^2}{\lambda}} \left(\frac{-2y\lambda}{\lambda^2} \right) \\ &= \begin{cases} \frac{2y}{\lambda} \cdot e^{-\frac{y^2}{\lambda}} & \text{si } y \geq 0 \\ 0 & \text{en o. c.} \end{cases} \end{aligned}$$

✓ Integra a 1.

Transformación simple

$$8. f_X(x) = \begin{cases} \frac{1}{2} \cdot e^{-|x|} & \text{si } -\infty < x < \infty. \\ 0 & \text{en o.c.} \end{cases}$$

$Y = \mu + \sigma X$, para $\sigma > 0$. Determine $f_Y(y)$.

Solución

$$Y = \mu + \sigma X.$$

$$Y - \mu = \sigma X.$$

$$\underbrace{\frac{Y - \mu}{\sigma}}_{h^{-1}(Y)} = X.$$

$$\begin{aligned} \frac{dh^{-1}}{dy} &= \frac{(1-0)\sigma - (Y-\mu)0}{\sigma^2} \\ &= \frac{\sigma}{\sigma^2} = \frac{1}{\sigma} \end{aligned}$$

$$\left| \frac{dh^{-1}}{dy} \right| = \frac{1}{\sigma}$$

$$f_Y(y) = f_X(h^{-1}(y)) \cdot \left| \frac{dh^{-1}}{dy} \right|$$

$$= \frac{1}{2} \cdot e^{-\left| \frac{Y - \mu}{\sigma} \right|} \cdot \frac{1}{\sigma}$$

$$= \begin{cases} \frac{1}{2\sigma} \cdot e^{-\left| \frac{Y - \mu}{\sigma} \right|} & -\infty < Y < \infty \\ 0 & \text{en o.c.} \end{cases}$$

$$Y = \mu + \sigma X$$

$$\text{si } X = -\infty$$

$$Y = \mu + \sigma \cdot -\infty$$

$$Y = -\infty$$

$$\text{si } X = \infty$$

$$Y = \mu + \sigma \cdot \infty$$

$$Y = \infty$$

9. $X \sim U\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Determine la $f_Y(y)$

si $Y = \tan(X)$.

$$f_X(x) = \begin{cases} \frac{1}{\frac{\pi}{2} - \left(-\frac{\pi}{2}\right) \rightarrow \pi} & \text{si } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ 0 & \text{en o. c.} \end{cases}$$

$$f_X(x) = \begin{cases} \frac{1}{\pi} & \text{si } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ 0 & \text{en o. c.} \end{cases}$$

Solución

$$Y = \tan(X).$$

$$\underbrace{\arctan(Y) = X}_{h^{-1}(y)}$$

$$\frac{dh^{-1}}{dy} = \frac{1}{1+y^2}$$

$$\left| \frac{dh^{-1}}{dy} \right| = \frac{1}{1+y^2}$$

$$f_Y(y) = f_X(h^{-1}(y)) \cdot \left| \frac{dh^{-1}}{dy} \right|$$

$$= \frac{1}{\pi} \cdot \frac{1}{1+y^2}$$

$$= \frac{1}{\pi(1+y^2)} \quad \left. \vphantom{\frac{1}{\pi(1+y^2)}} \right\} \begin{array}{l} \text{Llegamos} \\ \text{a una cauchy} \\ \text{estándar} \end{array}$$

$$\therefore Y \sim \text{Cau}(0, 1)$$

10.

a) $X \sim U(0,1)$. $Y = \tan(\pi(X - 1/2))$.

Encuentre la $F_Y(Y)$.

$$f_X(x) = \begin{cases} 1 & \text{si } 0 < x < 1 \\ 0 & \text{en o. c.} \end{cases}$$

Solution

$$F_Y(y) = P(Y \leq y) = P(\tan(\pi(X - 1/2)) \leq y)$$

$$= P(\pi(X - 1/2) \leq \arctan(y))$$

$$= P(X - 1/2 \leq \frac{\arctan(y)}{\pi})$$

$$= P(X \leq \frac{\arctan(y)}{\pi} + \frac{1}{2})$$

$$= \int_0^{\frac{\arctan(y)}{\pi} + \frac{1}{2}} 1 \, dx$$

$$= x \Big|_0^{\frac{\arctan(y)}{\pi} + \frac{1}{2}}$$

$$\therefore Y \sim \text{Cauchy} \quad \left(\mu=0, \gamma=1 \right) \quad \begin{cases} = \frac{\arctan(y)}{\pi} + \frac{1}{2} \\ = \frac{1}{\pi} \cdot \arctan(y) + \frac{1}{2} \end{cases}$$

10.

b) $X \sim \text{Can}(\overset{\mu}{0}, \overset{r}{1})$. $Y = rX + s$. con r y s constantes reales. Encuentre la $f_Y(y)$.

$$f_X(x) = \frac{1}{\pi(1+x^2)}$$

Solucion

$$Y = rX + s$$

$$Y - s = rX$$

$$\underbrace{\frac{Y-s}{r}}_{h^{-1}(y)} = X$$

$$\begin{aligned} \frac{dh^{-1}}{dy} &= \frac{(1-0)r - (y-s)0}{r^2} \\ &= \frac{r}{r^2} = \boxed{\frac{1}{r}} \end{aligned}$$

$$\left| \frac{dh^{-1}}{dy} \right| = \frac{1}{r}$$

$$f_Y(y) = f_X(h^{-1}(y)) \cdot \left| \frac{dh^{-1}}{dy} \right|$$

$$= \frac{1}{\pi(1 + (\frac{Y-s}{r})^2)} \cdot \frac{1}{r}$$

$$= \frac{1}{\pi r (1 + (\frac{Y-s}{r})^2)}$$

$$\therefore Y \sim \text{Can}(s, r)$$

$$11. X \sim \Gamma(\alpha, 1). \quad Y \sim \Gamma(\beta, 1).$$

$$u = \frac{X}{X+Y} \quad v = X+Y.$$

a) Determine la $f_{u,v}(u,v)$.

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \cdot x^{\alpha-1} \cdot y^{\beta-1} \cdot e^{-x} \cdot e^{-y} & \text{si } x \geq 0 \text{ y } y \geq 0 \\ 0 & \text{en o. c.} \end{cases}$$

Solucion:

$$u = \frac{X}{X+Y} \quad v = X+Y$$

$$v - Y = \textcircled{X}$$

$$uX + uY = X$$

$$uX - X = -uY$$

$$X(u-1) = -uY$$

$$X = \frac{-uY}{(u-1)}$$

$$\textcircled{X} = \frac{uY}{1-u}$$

$$\frac{uY}{1-u} = v - Y$$

$$\frac{uY}{1-u} + Y = v \Rightarrow Y \left(\frac{u}{1-u} + \frac{1}{1} \right) = v \Rightarrow Y \left(\frac{u+1-u}{1-u} \right) = v$$

$$\textcircled{Y} = \frac{v - uv}{1-u} \} g^{-1}(u,v)$$

$$u = \frac{x}{x+y}$$

$$v = x+y$$

$$v - x = \textcircled{y}$$

$$ux + uy = x$$

$$uy = x - ux$$

$$\textcircled{y} = \frac{x - ux}{u}$$

$$\frac{x - ux}{u} = v - x$$

$$x - ux = uv - ux$$

$$\textcircled{x} = uv$$

$$h^{-1}(u, v)$$

$$\left\{ \begin{array}{l} x = h^{-1} = uv \\ y = g^{-1} = v - uv \end{array} \right\}$$

$$J = \begin{vmatrix} \frac{dh^{-1}}{du} & \frac{dh^{-1}}{dv} \\ \frac{dg^{-1}}{du} & \frac{dg^{-1}}{dv} \end{vmatrix} = \begin{vmatrix} v & u \\ 1-v & 1-u \end{vmatrix}$$

$$J = v(1-u) - -v \cdot u$$

$$= v - uv + vu$$

$$= v$$

$$|J| = v$$

$$\begin{cases} x = h^{-1} = uv \\ y = g^{-1} = v - uv \end{cases}$$

$$f_{u,v}(u,v) = f_{x,y}(h^{-1}(u,v), g^{-1}(u,v)) \cdot |J|$$

$$= \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \cdot uv^{\alpha-1} v^{-uv}{}^{\beta-1} \cdot e^{-uv} \cdot e^{-(v-uv)} \cdot v$$

$$= \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \cdot uv^{\alpha-1} \cdot v^{-uv}{}^{\beta-1} \cdot e^{-uv-v+uv} \cdot v$$

$$= \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \cdot uv^{\alpha-1} \cdot v^{-uv}{}^{\beta-1} \cdot e^{-v} \cdot v$$

$$= \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \cdot uv^{\alpha-1} \cdot v(1-u)^{\beta-1} \cdot e^{-v} \cdot v$$

$$= \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \cdot u^{\alpha} v^{\alpha} \cdot u^{-1} v^{-1} \cdot v^{\beta} (1-u)^{\beta-1} \cdot v^{-1} \cdot (1-u) \cdot e^{-v} \cdot v$$

$$= \int \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \cdot u^{\alpha-1} \cdot v^{\alpha+\beta-1} (1-u)^{\beta-1} \cdot e^{-v}$$

$$s.t. \quad 0 < u < 1,$$

$$v > 0.$$

$$0 \text{ en o.c.}$$

$$\begin{aligned} x &\geq 0 \\ uv &\geq 0 \\ \boxed{u \geq 0} \end{aligned}$$

$$\begin{aligned} y &\geq 0 \\ v - uv &\geq 0 \\ v &\geq uv \\ \frac{v}{v} &> u \\ \boxed{1 > u} \end{aligned}$$

$$\begin{aligned} x &\geq 0 \\ uv &\geq 0 \\ \boxed{v \geq 0} \end{aligned} \quad \begin{aligned} y &\geq 0 \\ v(1-u) &\geq 0 \\ \boxed{v \geq 0} \end{aligned}$$

11.

b) Determine $f_u(u)$ y $f_v(v)$.

$$f_u(u) = \int_0^\infty \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \cdot u^{\alpha-1} v^{\alpha+\beta-1} (1-u)^{\beta-1} \cdot e^{-v} dv$$

$$= \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \cdot u^{\alpha-1} (1-u)^{\beta-1} \int_0^\infty v^{\alpha+\beta-1} \cdot e^{-v} dv$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{1}{\Gamma(\alpha+\beta)} \cdot u^{\alpha-1} (1-u)^{\beta-1} \int_0^\infty v^{\alpha+\beta-1} \cdot e^{-v} dv$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot u^{\alpha-1} (1-u)^{\beta-1} \underbrace{\int_0^\infty \frac{1}{\Gamma(\alpha+\beta)} \cdot v^{\alpha+\beta-1} \cdot e^{-v} dv}_{\Gamma(\alpha+\beta, 1)}$$

Beta(α, β)

$\left(\frac{1}{\beta^\alpha \Gamma(\alpha)} \cdot x^{\alpha-1} \cdot e^{-x/\beta} \right)$ Es un uno

$\therefore u \sim \text{Beta}(\alpha, \beta)$

$$f_V(v) = \int_0^1 \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \cdot u^{\alpha-1} v^{\alpha+\beta-1} (1-u)^{\beta-1} \cdot e^{-v} du$$

$$= \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \cdot v^{\alpha+\beta-1} \cdot e^{-v} \int_0^1 u^{\alpha-1} (1-u)^{\beta-1} du$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{1}{\Gamma(\alpha+\beta)} \cdot v^{\alpha+\beta-1} \cdot e^{-v} \int_0^1 u^{\alpha-1} (1-u)^{\beta-1} du$$

$$= \frac{1}{\Gamma(\alpha+\beta)} \cdot v^{\alpha+\beta-1} \cdot e^{-v} \int_0^1 \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot u^{\alpha-1} (1-u)^{\beta-1} du$$

$$= \underbrace{\frac{1}{\Gamma(\alpha+\beta)} \cdot v^{\alpha+\beta-1} \cdot e^{-v}}_{\Gamma(\alpha+\beta, 1)} \underbrace{\int_0^1 \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot u^{\alpha-1} (1-u)^{\beta-1} du}_{\text{Beta}(\alpha, \beta) \rightarrow \text{ES un}}_{\substack{\text{si } 0 < x < 1 \\ 0 \text{ en o. c.}}}$$

$$\left(\frac{1}{\beta^\alpha \Gamma(\alpha)} \cdot x^{\alpha-1} \cdot e^{-x/\beta} \right)$$

$$\therefore V \sim \Gamma(\alpha+\beta, 1)$$

$$12. X \sim U(0,1).$$

$$f_Y(y) = \begin{cases} 2y & \text{si } 0 < y < 1 \\ 0 & \text{en o.c.} \end{cases} \quad \left. \begin{array}{l} u=0.0 \\ u=0.1 \\ u=1.0 \\ u=1.1 \end{array} \right\} 0 < u < 1$$

son independientes.

a) $u = XY$. Determine la $f_u(u)$.

$$f_{X,Y}(x,y) = \begin{cases} 2y & \text{si } 0 < y < 1 \text{ y } 0 < x < 1 \\ 0 & \text{en o.c.} \end{cases}$$

Solucion

$$u = XY$$

$$\underbrace{\frac{u}{x}}_{h^{-1}(u)} = Y$$

$$\begin{aligned} \frac{dh^{-1}}{du} &= \frac{1 \cdot x - u \cdot 0}{x^2} \\ &= \frac{x}{x^2} = \frac{1}{x} \end{aligned}$$

$$\left| \frac{dh^{-1}}{du} \right| = \frac{1}{x}$$

$$g(x,u) = f_{X,Y}(x, h^{-1}(u)) \cdot \left| \frac{dh^{-1}}{du} \right|$$

$$= 2 \cdot \frac{u}{x} \cdot \frac{1}{x}$$

$$0 < y < 1$$

$$0 < \frac{u}{x} < 1$$

$$= \begin{cases} \frac{2u}{x^2} & \text{si } 0 < x < 1, 0 < u < 1, \\ & \text{y } 0 < u < x < 1 \\ 0 & \text{en o.c.} \end{cases}$$

$$f_u(u) =$$

$$= \int_u^1 \frac{2u}{x^2} dx$$

$$= 2u \int_u^1 \frac{1}{x^2} dx$$

$$= 2u \int_u^1 x^{-2} dx$$

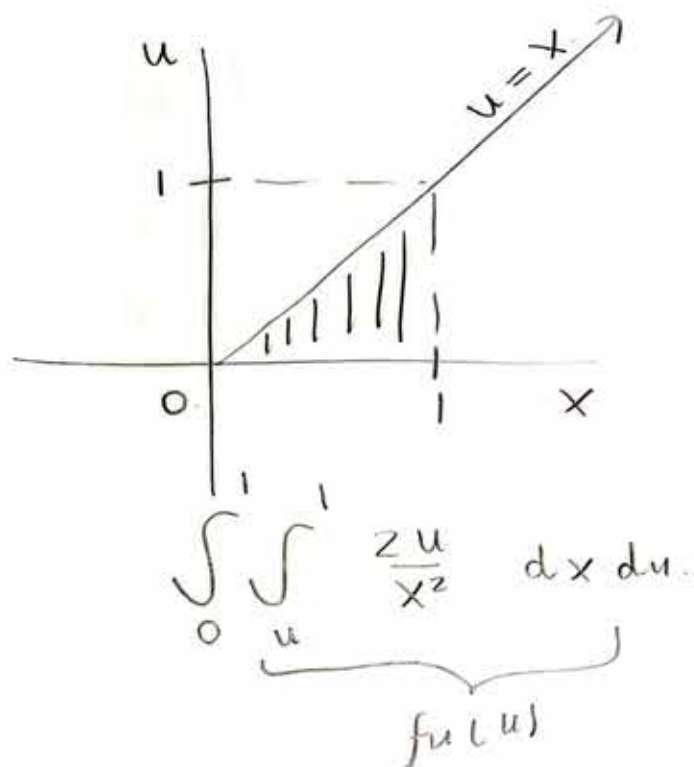
$$= 2u \left(-\frac{1}{x} \Big|_u^1 \right)$$

$$= 2u \left(-1 + \frac{1}{u} \right)$$

$$= -2u + \frac{2u}{u}$$

$$= -2u + 2$$

$$= \begin{cases} 2-2u & \text{si } 0 < u < 1 \\ 0 & \text{en } 0, c \end{cases}$$



✓ Integra a 1.

b) $u = \frac{x}{y}$. Determina la $f_u(u)$.

$$f_{x,y}(x,y) = \begin{cases} 2y & \text{si } 0 < y < 1, 0 < x < 1 \\ 0 & \text{en o.c.} \end{cases}$$

$$\begin{aligned} u = \frac{x}{y} &= \frac{0}{0} \\ &= \frac{0}{1} = 0 \\ &= \frac{1}{0} = \infty \\ &= \frac{1}{1} = 1. \end{aligned}$$

Solucion

si $u < 0$

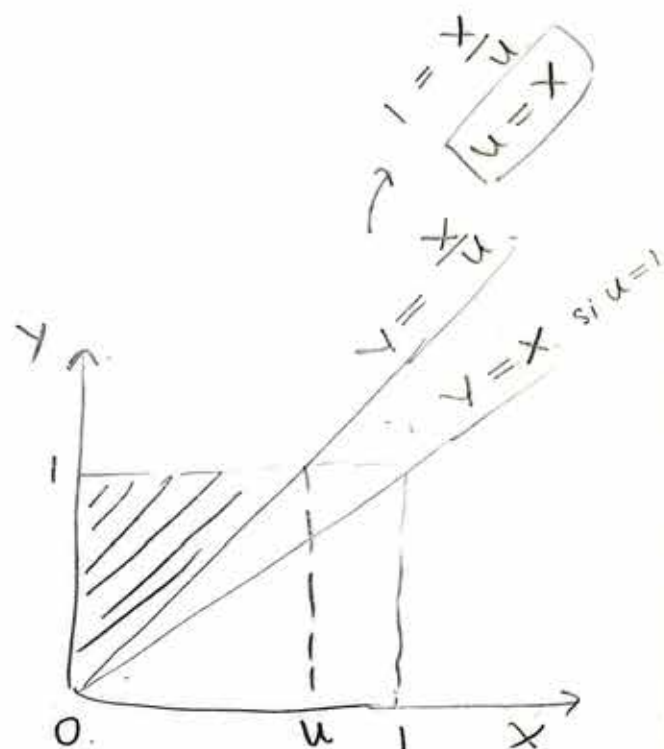
$$\Rightarrow F_u(u) = 0$$

si $0 < u < 1$.

$$\begin{aligned} \Rightarrow F_u(u) &= P(U \leq u) = P\left(\frac{x}{y} \leq u\right) \\ &= P(x \leq uy) \\ &= P\left(\frac{x}{u} \leq y\right) \end{aligned}$$

$$= \int_0^u \int_{\frac{x}{u}}^1 2y \, dy \, dx$$

$$= \int_0^u \left[y^2 \right]_{\frac{x}{u}}^1 \, dx = \int_0^u \left(1 - \frac{x^2}{u^2} \right) \, dx$$



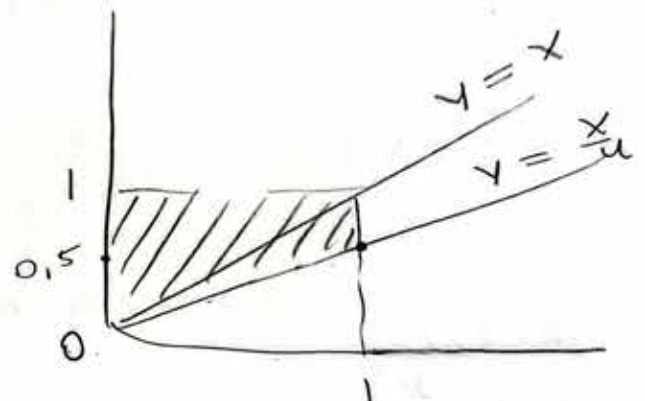
$$\begin{aligned}
 &= \int_0^u \left(1 - \frac{x^2}{u^2} \right) dx \\
 &= \int_0^u \left(1 - \frac{1}{u^2} \cdot x^2 \right) dx \\
 &= \left(x - \frac{1}{u^2} \cdot \frac{x^3}{3} \right) \Big|_0^u \\
 &= u - \frac{1}{u^2} \cdot \frac{u^3}{3} \\
 &= u - \frac{u^3}{3u^2} \\
 &= \boxed{u - \frac{u}{3}}
 \end{aligned}$$

$$0,5 = \frac{1}{u}$$

$$u = 2$$

$$y = \frac{x}{u}$$

$$z = x$$



$$y = \frac{x}{u}$$

$$0,5 = \frac{1}{u}$$

$$u = \frac{1}{0,5} = 2$$

$$\begin{aligned}
 &\text{Si } 1 < u < \infty \\
 &= \int_0^1 \int_{\frac{x}{u}}^1 2y \, dy \, dx \\
 &= \int_0^1 \left(2 \cdot \frac{y^2}{2} \right) \Big|_{\frac{x}{u}}^1 dx \\
 &= \int_0^1 \left(1 - \frac{x^2}{u^2} \right) dx
 \end{aligned}$$

$$= x - \frac{1}{u^2} \cdot \frac{x^3}{3} \Big|_0^1$$

$$= 1 - \frac{1}{3u^2}$$

$$\therefore F_u(u) = \begin{cases} 0 & \text{si } u < 0 \\ u - \frac{u}{3} & \text{si } 0 < u < 1 \\ 1 - \frac{1}{3u^2} & \text{si } 1 < u < \infty \end{cases}$$

$$f_u(u) = \begin{cases} 1 - \frac{1}{3} & \text{si } 0 < u < 1 \\ 0 - \left[\frac{0(3u^2) - 1(6u)}{9u^4} \right] \end{cases}$$

$$\frac{6u}{9u^4} = \frac{2}{3u^3} \quad \text{si } 1 < u < \infty$$

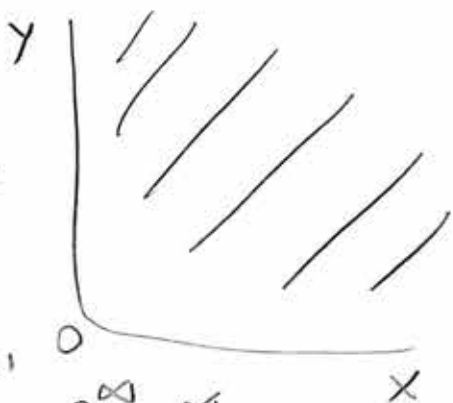
$$f_u(u) = \begin{cases} 1 - \frac{1}{3} & \text{si } 0 < u < 1 \\ \frac{2}{3u^3} & \text{si } 1 < u < \infty \\ 0 & \text{en o.c} \end{cases}$$

✓ Integra a 1.

$$13. f_{X,Y}(x,y) = \begin{cases} c(x+y)e^{-x-y} & \text{si } x > 0, y > 0 \\ 0 & \text{en o. c.} \end{cases}$$

a) Determina el valor de la constante "C".

$$\Rightarrow \int_0^{\infty} \int_0^{\infty} c(x+y)e^{-x-y} dy dx = 1$$



$$\Rightarrow c \int_0^{\infty} \int_0^{\infty} x \cdot e^{-x-y} + y e^{-x-y} dy dx = 1$$

$$\int_0^{\infty} \int_0^{\infty} f_{X,Y}(x,y) dy dx = 1$$

$$\Rightarrow c \left[\int_0^{\infty} \int_0^{\infty} x \cdot e^{-x-y} dy dx + \int_0^{\infty} \int_0^{\infty} y e^{-x-y} dy dx \right] = 1$$

$$\begin{matrix} u=y & dv=e^{-y} \\ du=dy & v=-e^{-y} \end{matrix}$$

$$\Rightarrow c \left[\int_0^{\infty} x e^{-x} \int_0^{\infty} e^{-y} dy dx + \int_0^{\infty} e^{-x} \int_0^{\infty} y \cdot e^{-y} dy dx \right] = 1$$

$$\Rightarrow c \left[\int_0^{\infty} x e^{-x} (0+1) dx + \int_0^{\infty} e^{-x} (-y e^{-y} - \int e^{-y} dy) \Big|_0^{\infty} dx \right] = 1$$

$$\begin{matrix} u=x & dv=e^{-x} \\ du=dx & v=-e^{-x} \end{matrix}$$

$$\Rightarrow c \left[-x e^{-x} - \int e^{-x} dx \Big|_0^{\infty} + \int_0^{\infty} e^{-x} (-y e^{-y} - e^{-y} \Big|_0^{\infty}) dx \right] = 1$$

$$\Rightarrow c \left[-x e^{-x} - e^{-x} \Big|_0^{\infty} + \int_0^{\infty} e^{-x} (0 - (-1)) dx \right] = 1$$

$$\Rightarrow c \left[0 - (-1) + \int_0^{\infty} e^{-x} dx \right] = 1$$

$$\Rightarrow c \left[1 + (-e^{-x} |_0^{\infty}) \right] = 1$$

$$\Rightarrow c \left[1 + (0 + 1) \right] = 1$$

$$\Rightarrow c \left[1 + 1 \right] = 1$$

$$\Rightarrow c \cdot 2 = 1$$

$$\Rightarrow 2c = 1$$

$$\boxed{\Rightarrow c = \frac{1}{2}}$$

13. b) $u = x + y$. Determine la $f_u(u)$.

$$f_{x,y}(x,y) = \begin{cases} \frac{1}{2} (x+y) e^{-x-y} & \text{si } x > 0, y > 0 \\ 0 & \text{en o.c.} \end{cases}$$

Solucion:

$$u = x + y$$

$$\underbrace{u - x = y}_{h^{-1}(u)}$$

$$\frac{dh^{-1}}{du} = 1$$

$$\left| \frac{dh^{-1}}{du} \right| = 1$$

$$g(x,u) = f_{x,y}(x, h^{-1}(u)) \cdot \left| \frac{dh^{-1}}{du} \right|$$

$$= \frac{1}{2} (x + u - x) e^{-x - (u-x)}$$

$$= \frac{1}{2} u \cdot e^{-u} \quad \text{si } \begin{matrix} x > 0, u > 0 \\ y > u - x \end{matrix} \quad \begin{matrix} y > 0 \\ u - x > y \\ u > x \end{matrix}$$

$$u = x + y$$

$$u = 0 + 0$$

$$u = 0 + \infty$$

$$u = \infty + 0$$

$$u = \infty + \infty$$

$$\underbrace{\hspace{10em}}_{u > 0}$$

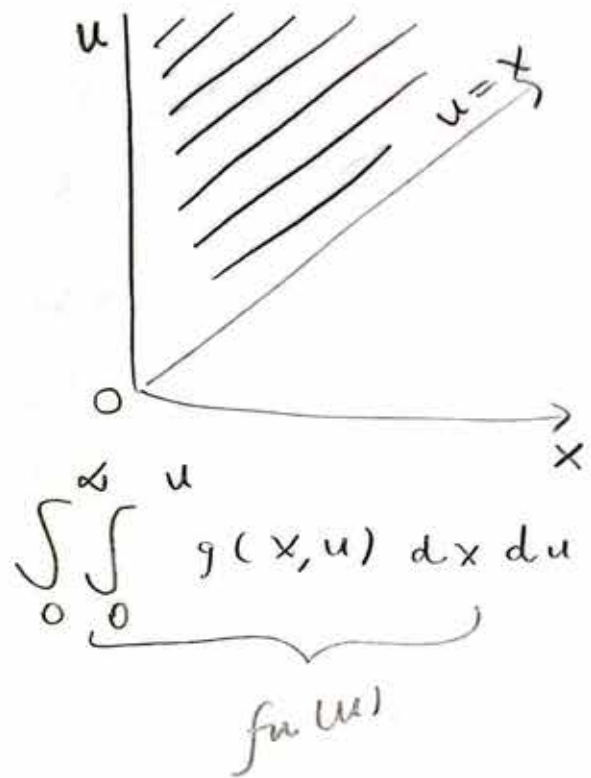
$$f_u(u) =$$

$$= \int_0^u \frac{1}{2} \cdot u \cdot e^{-u} dx$$

$$= \frac{1}{2} \cdot u \cdot e^{-u} \int_0^u 1 dx$$

$$= \begin{cases} \frac{1}{2} \cdot u^2 \cdot e^{-u} & \text{si } u > 0 \\ 0 & \text{en o. c.} \end{cases}$$

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a 1.



$$14. f_{x,y}(x,y) = \begin{cases} cxy & \text{si } 0 < x < y < 1 \\ 0 & \text{en o. c.} \end{cases}$$

a) Determine el valor de la constante "c".

$$\Rightarrow \int_0^1 \int_0^y cxy \, dx \, dy = 1$$

$$\Rightarrow c \int_0^1 y \int_0^y x \, dx \, dy = 1$$

$$\Rightarrow c \int_0^1 y \cdot \left(\frac{x^2}{2} \Big|_0^y \right) dy = 1$$

$$\Rightarrow c \int_0^1 \frac{y^3}{2} \, dy = 1$$

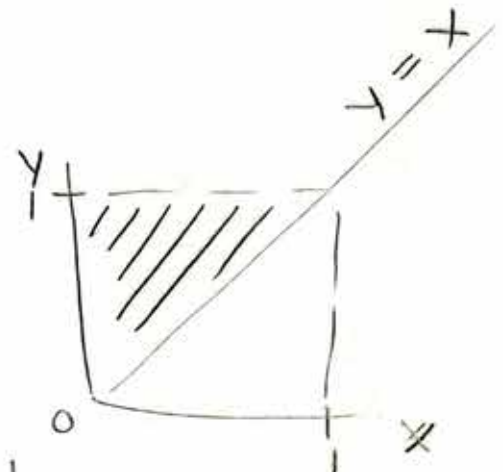
$$\Rightarrow \frac{c}{2} \int_0^1 y^3 \, dy = 1$$

$$\Rightarrow \frac{c}{2} \cdot \left(\frac{y^4}{4} \Big|_0^1 \right) = 1$$

$$\Rightarrow \frac{c}{2} \left(\frac{1}{4} \right) = 1$$

$$\Rightarrow \frac{c}{8} = 1$$

$$\boxed{\Rightarrow c = 8}$$



$$\int_0^1 \int_0^y cxy \, dx \, dy = 1$$

14. b). $U = XY$. Determine la $f_U(u)$.

$$f_{X,Y}(X,Y) = \begin{cases} 8XY & \text{si } 0 < X < Y < 1 \\ 0 & \text{en o. c.} \end{cases}$$

Solucion

$$u = XY$$

$$\underbrace{\frac{u}{Y} = X}_{h^{-1}(u)}$$

$$\frac{dh^{-1}}{du} = \frac{1 \cdot Y - u \cdot 0}{Y^2}$$

$$= \frac{Y}{Y^2} = \frac{1}{Y}$$

$$\left| \frac{dh^{-1}}{du} \right| = \frac{1}{Y}$$

$$g(u,Y) = f_{X,Y}(h^{-1}(u), Y) \cdot \left| \frac{dh^{-1}}{du} \right|$$

$$= 8 \cdot \frac{u}{Y} \cdot Y \cdot \frac{1}{Y}$$

$$0 < X < Y < 1$$

$$0 < \frac{u}{Y} < Y < 1$$

$$= \begin{cases} \frac{8u}{Y} & 0 < u < Y^2 < 1 \\ 0 & \text{en o. c.} \end{cases}$$

$$0 < u < Y^2 < 1$$

$$Y > \sqrt{u}$$

$$f(u).$$

$$= \int_{\sqrt{u}}^1 \frac{8u}{y} dy$$

$$= 8u \int_{\sqrt{u}}^1 \frac{1}{y} dy$$

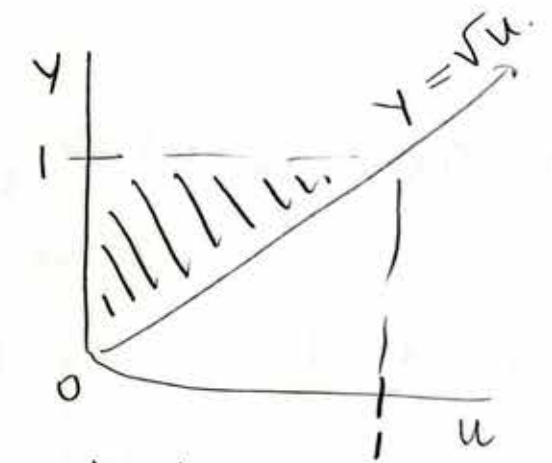
$$= 8u (\ln y \big|_{\sqrt{u}}^1)$$

$$= 8u (0 - \ln \sqrt{u}).$$

$$= -8u \cdot \ln u^{\frac{1}{2}}.$$

$$= -8u \cdot \frac{1}{2} \cdot \ln u$$

$$= \begin{cases} -4u \cdot \ln u & \text{si } 0 < u < 1 \\ 0 & \text{en o. c.} \end{cases}$$



$$\int_0^1 \int_{\sqrt{u}}^1 \frac{8u}{y} dy du$$

$$f(u).$$

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$$15. f_{x,y}(x,y) = \begin{cases} 2e^{-x-2y} & \text{si } x > 0, y > 0 \\ 0 & \text{en o.c.} \end{cases}$$

a) $u = x + y$. Determine la $f_u(u)$.

Solucion

$$u = x + y$$

$$\underbrace{u - x = y}_{h^{-1}(u)}$$

$$h^{-1}(u)$$

$$\frac{dh^{-1}}{du} = 1$$

$$\left| \frac{dh^{-1}}{du} \right| = 1.$$

$$g(x,u) = f_{x,y}(x, h^{-1}(u)) \cdot \left| \frac{dh^{-1}}{du} \right|$$

$$= 2e^{-x-2(u-x)}$$

$$= \begin{cases} 2e^{-x-2u+2x} & \text{si } \begin{matrix} x > 0, \\ u > 0, \\ u > x. \end{matrix} \\ 0 & \text{en o.c.} \end{cases}$$

$$y > 0.$$

$$u - x > 0.$$

$$u > x.$$

$$f_u(u) =$$

$$= \int_0^u 2e^{-x+2x} \cdot e^{-2u} dx$$

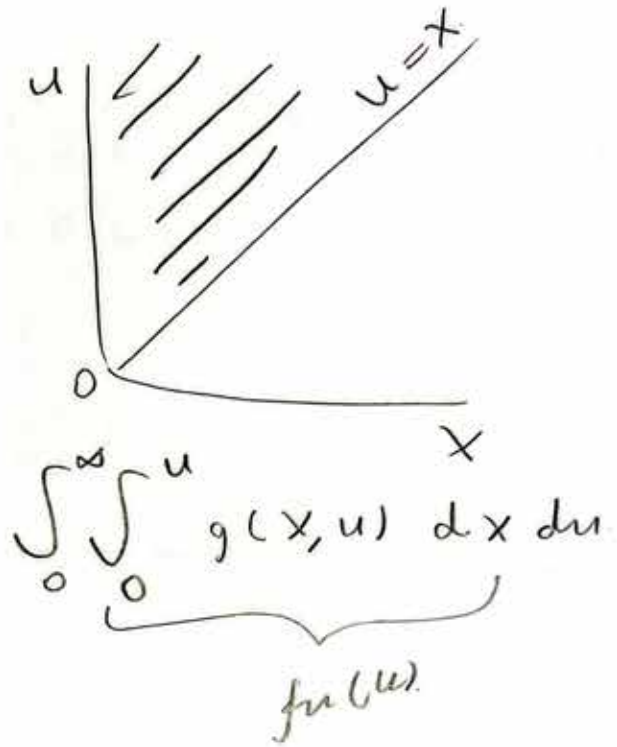
$$= 2e^{-2u} \int_0^u e^x dx$$

$$= 2e^{-2u} (e^x \big|_0^u)$$

$$= 2e^{-2u} (e^u - 1)$$

$$= 2e^{-2u+u} - 2e^{-2u}$$

$$= \begin{cases} 2e^{-u} - 2e^{-2u} & \text{si } u > 0 \\ 0 & \text{en o.c.} \end{cases}$$



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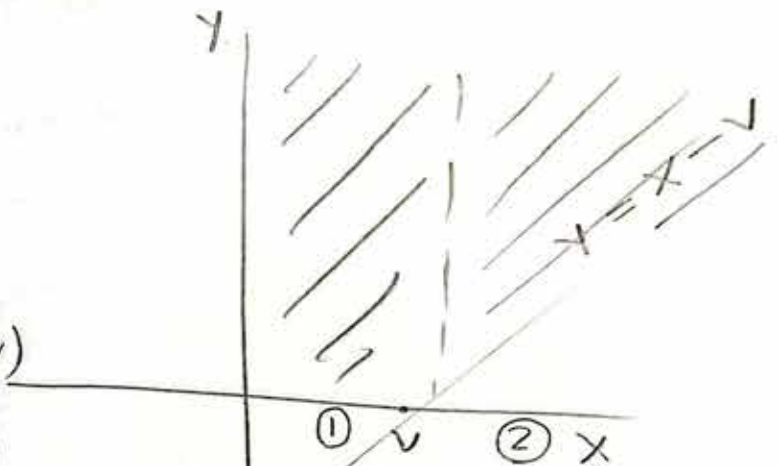
$$15. \quad b) \quad f_{X,Y}(x,y) = \begin{cases} 2e^{-x-2y} & \text{si } x > 0, y > 0 \\ 0 & \text{en o. c.} \end{cases}$$

$$V = X - Y.$$

$$\left. \begin{aligned} V = X - Y &= 0 - 0 = 0. \\ &= 0 - \infty = -\infty \\ &= \infty - 0 = \infty. \end{aligned} \right\} \begin{aligned} &-\infty < V < \infty. \\ &y > 0, \\ &x > 0 \end{aligned}$$

$$\text{Si } V > 0.$$

$$\begin{aligned} \Rightarrow F_V(V) &= P(V \leq v) \\ &= P(X - Y \leq v) \\ &= P(X - v \leq Y) \end{aligned}$$



$$\begin{aligned} &= \int_0^v \int_0^\infty 2e^{-x-2y} dy dx + \int_v^\infty \int_{x-v}^\infty 2e^{-x-2y} dy dx \\ &= 2 \int_0^v e^{-x} \int_0^\infty e^{-2y} dy dx + 2 \int_v^\infty e^{-x} \int_{x-v}^\infty e^{-2y} dy dx \\ &= 2 \int_0^v e^{-x} \left(-\frac{e^{-2y}}{2} \Big|_0^\infty \right) dx + 2 \int_v^\infty e^{-x} \left(-\frac{e^{-2y}}{2} \Big|_{x-v}^\infty \right) dx \end{aligned}$$

$$\begin{aligned}
&= \int_0^v e^{-x} (0+1) + \int_v^\infty e^{-x} (0 + e^{-2(x-v)}) dx \\
&= -e^{-x} \Big|_0^v + \int_v^\infty e^{-x} \cdot e^{-2x+2v} dx \\
&= -e^{-v} + 1 + e^{2v} \int_v^\infty e^{-3x} dx \\
&= -e^{-v} + 1 + e^{2v} \left(-\frac{e^{-3x}}{3} \Big|_v^\infty \right) \\
&= -e^{-v} + 1 + \frac{e^{2v}}{3} (0 + e^{-3v}) \\
&= -e^{-v} + 1 + \frac{e^{2v-3v}}{3} \\
&= -e^{-v} + 1 + \frac{e^{-v}}{3}
\end{aligned}$$

$$\text{If } v < 0.$$

$$\Rightarrow F_V(v) = P(V \leq v)$$

$$= P(X - Y \leq v)$$

$$= P(X - v \leq Y)$$

$$= \int_0^{\infty} \int_{x-v}^{\infty} 2e^{-x-2y} dy dx.$$

$$= 2 \int_0^{\infty} e^{-x} \int_{x-v}^{\infty} e^{-2y} dy dx$$

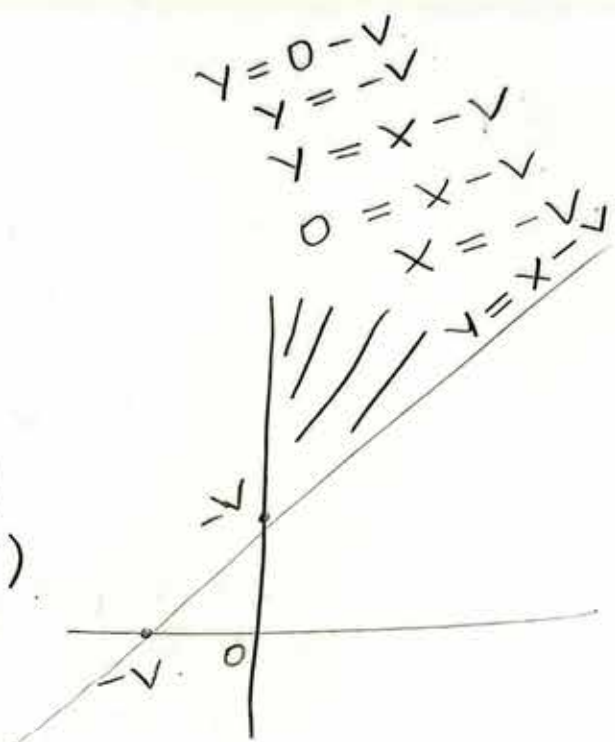
$$= 2 \int_0^{\infty} e^{-x} \left(-\frac{e^{-2y}}{2} \Big|_{x-v}^{\infty} \right) dx.$$

$$= \int_0^{\infty} e^{-x} (0 + e^{-2x+2v}) dx.$$

$$= e^{2v} \int_0^{\infty} e^{-3x} dx$$

$$= e^{\frac{2v}{3}} \left(-e^{-3x} \Big|_0^{\infty} \right)$$

$$= e^{\frac{2v}{3}} (0 + 1) = e^{\frac{2v}{3}}$$



$$\therefore F_V(v) = \begin{cases} -e^{-v} + 1 + \frac{e^{-v}}{3} & \text{si } v > 0 \\ \frac{e^{2v}}{3} & \text{si } v < 0 \end{cases}$$

$$f_V(v) = \frac{dF_V(v)}{dv} = e^{-v} + \frac{1}{3} \cdot e^{-v} \cdot -1$$

$$= e^{-v} - \frac{e^{-v}}{3}$$

$$= \frac{1}{3} \cdot e^{2v} \cdot 2$$

$$= \frac{2}{3} e^{2v}$$

$$f_V(v) = \begin{cases} e^{-v} - \frac{e^{-v}}{3} & \text{si } v > 0. \\ \frac{2}{3} e^{2v} & \text{si } v < 0. \end{cases}$$

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$$17. X \sim U(0, 1). Y \sim \text{Exp}(\lambda).$$

$U = X + Y$. Determine $f_U(u)$.

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{\lambda} \cdot e^{-y/\lambda} & \text{if } 0 < x < 1, y \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

Solution

$$u = x + y$$

$$u - x = y$$

$$h^{-1}(u).$$

$$\frac{dh^{-1}}{du} = 1$$

$$\left| \frac{dh^{-1}}{du} \right| = 1$$

$$g(x,u) = f_{X,Y}(x, h^{-1}(u)) \cdot \left| \frac{dh^{-1}}{du} \right|$$

$$= \begin{cases} \frac{1}{\lambda} \cdot e^{-\frac{(u-x)}{\lambda}} & \text{if } 0 < x < 1, y \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

$u \geq x,$
 $u > 0,$
 $0 < u < 1$

$u - x \geq 0$
 $u \geq x$

$$\left. \begin{aligned} u = x + y &= 0 + 0 = 0 \\ &= 0 + \infty = \infty \\ &= 1 + 0 = 1 \\ &= 1 + \infty = \infty \end{aligned} \right\} \begin{aligned} 0 < u < 1 \\ u > 0. \end{aligned}$$

$$f(u) \text{ for } 0 < u < 1$$

$$= \int_0^u \frac{1}{\lambda} \cdot e^{\frac{x-u}{\lambda}} dx.$$

$$= \frac{1}{\lambda} \int_0^u e^{\frac{x}{\lambda}} \cdot e^{-\frac{u}{\lambda}} dx.$$

$$= \frac{1}{\lambda} e^{-u/\lambda} \int_0^u e^{\frac{x}{\lambda}} dx$$

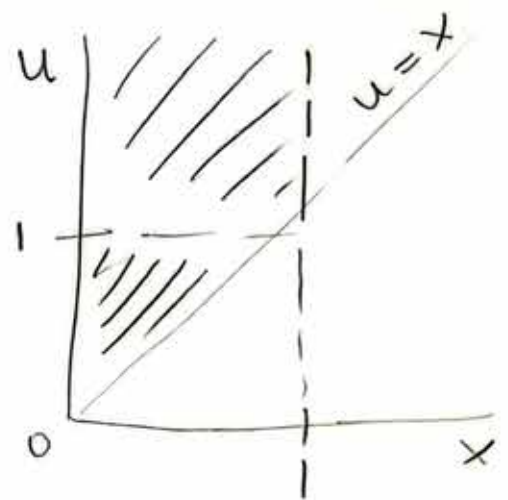
$$= \frac{1}{\lambda} e^{-u/\lambda} \left(\frac{e^{x/\lambda}}{\frac{1}{\lambda}} \bigg|_0^u \right)$$

$$= \frac{1}{\lambda} e^{-u/\lambda} \left(\lambda e^{x/\lambda} \bigg|_0^u \right)$$

$$= \frac{1}{\lambda} e^{-u/\lambda} \left(\lambda e^{u/\lambda} - \lambda \right)$$

$$= e^{-u/\lambda + u/\lambda} - e^{-u/\lambda}.$$

$$= 1 - e^{-u/\lambda}$$



$$\int_0^1 \int_0^u g(x, u) dx du.$$

$$f(u) \text{ for } 0 < u < 1$$

$$\int_1^\infty \int_0^1 g(x, u) dx du.$$

$$f(u) \text{ for } u > 1$$

$$f_u(u) \text{ si } u > 1$$

$$= \int_0^1 \frac{1}{\lambda} \cdot e^{x \frac{-u}{\lambda}} dx$$

$$= \frac{1}{\lambda} \cdot e^{-u/\lambda} \int_0^1 e^{x/\lambda} dx$$

$$= \frac{1}{\lambda} \cdot e^{-u/\lambda} \left(\lambda e^{x/\lambda} \Big|_0^1 \right)$$

$$= \frac{1}{\lambda} e^{-u/\lambda} \left(\lambda e^{1/\lambda} - \lambda \right)$$

$$= e^{-u/\lambda + 1/\lambda} - e^{-u/\lambda}$$

$$= e^{-\frac{u+1}{\lambda}} - e^{-u/\lambda} \text{ si } u > 1$$

$$\therefore f_u(u) = \begin{cases} 1 - e^{-u/\lambda} & \text{si } 0 < u < 1 \\ e^{-\frac{u+1}{\lambda}} - e^{-u/\lambda} & \text{si } u > 1 \\ 0 & \text{en o.c} \end{cases}$$

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$$19. f_{x,y}(x,y) = \begin{cases} 6e^{-3x-2y} & \text{en } x \geq 0, y \geq 0 \\ 0 & \text{en o.c.} \end{cases}$$

$u = x + y$. Determine la f.u.

Solución

$$\begin{aligned} u = x + y &= 0 + 0 = 0 \\ &= 0 + \infty = \infty \\ &= \infty + \infty = \infty \\ &= \infty + 0 = \infty \end{aligned} \left\{ \begin{array}{l} u \geq 0 \end{array} \right.$$

$$\frac{dh^{-1}}{du} = 1$$

$$\left| \frac{dh^{-1}}{du} \right| = 1$$

$$u = x + y$$

$$\underbrace{u - x = y}_{h^{-1}(u)}$$

$$g(x,u) = f_{x,y}(x, h^{-1}(u)) \cdot \left| \frac{dh^{-1}}{du} \right|$$

$$= 6e^{-3x-2(u-x)}$$

$$= 6e^{-3x-2u+2x}$$

$$= \begin{cases} 6e^{-x-2u} & x \geq 0 \\ 0 & \text{en o.c.} \end{cases}$$

$$y \geq 0$$

$$u - x \geq 0$$

$$u \geq x$$

$$u \geq 0$$

$$u \geq x$$

$$f_u(u)$$

$$= \int_0^u b e^{-x-2u} dx$$

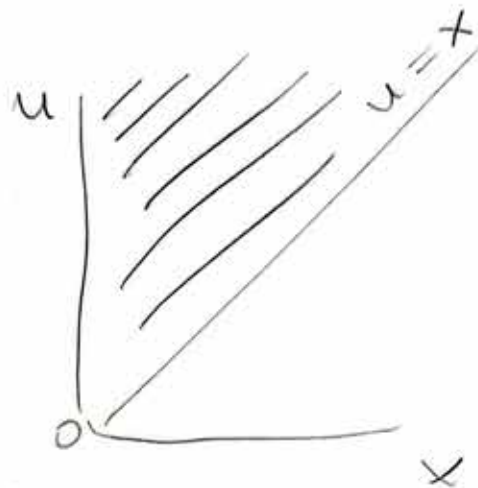
$$= b \cdot e^{-2u} \int_0^u e^{-x} dx$$

$$= b \cdot e^{-2u} (-e^{-x} \Big|_0^u)$$

$$= b e^{-2u} (-e^{-u} + 1)$$

$$= -b e^{-2u-u} + b e^{-2u}$$

$$= \begin{cases} -b e^{-3u} + b e^{-2u} & \text{si } u \geq 0 \\ 0 & \text{en o.c} \end{cases}$$



$$\int_0^{\infty} \underbrace{\int_0^u g(x,u) dx}_{f_u(u)} du$$

$$f_u(u)$$

si $u \geq 0$

✓ si integra a 1.

$$20. \quad f_{X,Y}(x,y) = \begin{cases} 5xe^{-xy} & \text{si } 0.2 \leq x \leq 0.4, \\ 0 & \text{en o.c.} \end{cases} \quad y \geq 0.$$

$u = XY$. Determine la f.d. de u .

Solución

$$u = XY$$

$$\underbrace{\frac{u}{x}} = Y$$

$$h^{-1}(u).$$

$$\begin{aligned} \frac{dh^{-1}}{du} &= \frac{1 \cdot x - u \cdot 0}{x^2} \\ &= \frac{x}{x^2} = \frac{1}{x} \end{aligned}$$

$$\left| \frac{dh^{-1}}{du} \right| = \frac{1}{x}$$

$$g(x,u) = f_{X,Y}(x, h^{-1}(u)) \cdot \left| \frac{dh^{-1}}{du} \right|$$

$$= 5x e^{-x(\frac{u}{x})} \cdot \frac{1}{x}$$

$$= \begin{cases} 5e^{-u} & \text{si } 0.2 < x < 0.4 \\ 0 & \text{en o.c.} \end{cases} \quad u > 0$$

$$y \geq 0.$$

$$\frac{u}{x} \geq 0$$

$$u > 0$$

$$f_u(u)$$

$$= \int_{0,2}^{0,4} 5e^{-u} dx$$

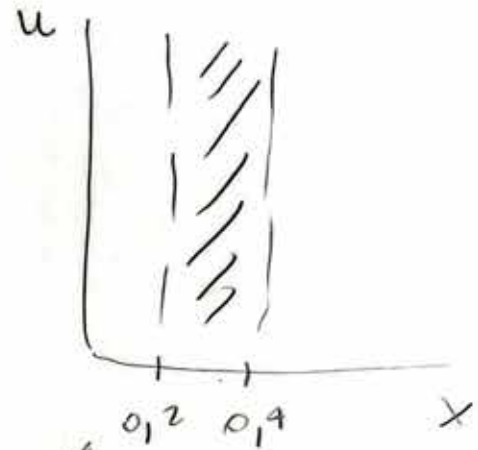
$$= 5e^{-u} \int_{0,2}^{0,4} 1 dx$$

$$= 5e^{-u} (x \Big|_{0,2}^{0,4})$$

$$= 5e^{-u} (0,4 - 0,2)$$

$$= 5e^{-u} \cdot \frac{1}{5}$$

$$= \begin{cases} e^{-u} & \text{si } u \geq 0 \\ 0 & \text{en o.c.} \end{cases}$$



$$\int_0^{\infty} \underbrace{\int_{0,2}^{0,4} 5e^{-u} dx}_{f_u(u)} du$$

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21.

$$f_{X,Y}(x,y) = \begin{cases} e^{-x-y} & \text{si } x \geq 0, y \geq 0 \\ 0 & \text{en o. c.} \end{cases}$$

$u = \frac{x}{y}$. $v = x + y$. Determine la $f_{u,v}(u,v)$.

Solution

$$u = \frac{x}{y} \quad v = x + y$$

$$uy = \textcircled{x} \quad v - y = \textcircled{y}$$

$$\downarrow \quad \downarrow$$

$$uy = v - y$$

$$uy + y = v$$

$$y(u+1) = v$$

$$\textcircled{y} = \frac{v}{u+1}$$

$$\underbrace{\hspace{10em}}_{g^{-1}(u,v)}$$

$$u = \frac{x}{y} \quad v = x + y$$

$$uy = x \quad v - x = \textcircled{y}$$

$$\textcircled{y} = \frac{x}{u}$$

$$\downarrow \quad \downarrow$$

$$\frac{x}{u} = v - x$$

$$\frac{x}{u} + x = v$$

$$x\left(\frac{1}{u} + 1\right) = v$$

$$x\left(\frac{1+u}{u}\right) = v$$

$$\textcircled{x} = \frac{vu}{1+u}$$

$$\underbrace{\hspace{10em}}_{h^{-1}(u,v)}$$

$$|J| = \begin{vmatrix} \frac{dh^{-1}}{du} & \frac{dh^{-1}}{dv} \\ \frac{dg^{-1}}{du} & \frac{dg^{-1}}{dv} \end{vmatrix} \quad \begin{cases} y = g^{-1} = \frac{v}{u+1} \\ x = h^{-1} = \frac{vu}{1+u} \end{cases}$$

$$|J| = \begin{vmatrix} \frac{v(1+u) - vu(1)}{(1+u)^2} & \frac{u(1+u) - \cancel{vu(0)}}{(1+u)^2} \\ \frac{\cancel{0(u+1)} - v(1)}{(u+1)^2} & \frac{1(u+1) - \cancel{v(0)}}{(u+1)^2} \end{vmatrix}$$

$$|J| = \begin{vmatrix} \frac{v + u\cancel{v} - vu}{(1+u)^2} & \frac{u}{(1+u)} \\ \frac{-v}{(u+1)^2} & \frac{1}{(u+1)} \end{vmatrix}$$

$$|J| = \frac{v}{(1+u)^3} + \frac{vu}{(u+1)^3} = \frac{v + uv}{(u+1)^3} = \frac{v(1+u)}{(u+1)^3} = \frac{v}{(u+1)^2}$$

$$f_{u,v}(u,v) = f_{x,y}(h^{-1}(u,v), g^{-1}(u,v)) \cdot |J|$$

$$= e^{-\frac{vu}{1+u} - \frac{v}{u+1}} \cdot \frac{v}{(u+1)^2}$$

$$= e^{-\frac{vu+v}{1+u}} \cdot \frac{v}{(u+1)^2}$$

$$= e^{-\frac{v(u+1)}{1+u}} \cdot \frac{v}{(u+1)^2}$$

$$= \begin{cases} e^{-v} \cdot \frac{v}{(u+1)^2} & \text{si } u \geq 0 \\ & \text{y } v \geq 0 \\ 0 & \text{en o. c.} \end{cases}$$

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$$x \geq 0$$

$$\frac{vu}{1+u} \geq 0$$

$$vu \geq 0$$

$$\boxed{u \geq 0}$$

$$v \geq 0$$

$$\frac{v}{u+1} \geq 0$$

$$\boxed{v \geq 0}$$

$$f_u(u)$$

$$= \int_0^{\infty} e^{-v} \cdot \frac{v}{(u+1)^2} dv.$$

$$= \frac{1}{(u+1)^2} \int_0^{\infty} v \cdot e^{-v} dv.$$

$u=v \quad dw=e^{-v}$
 $du=dv \quad w=-e^{-v}$

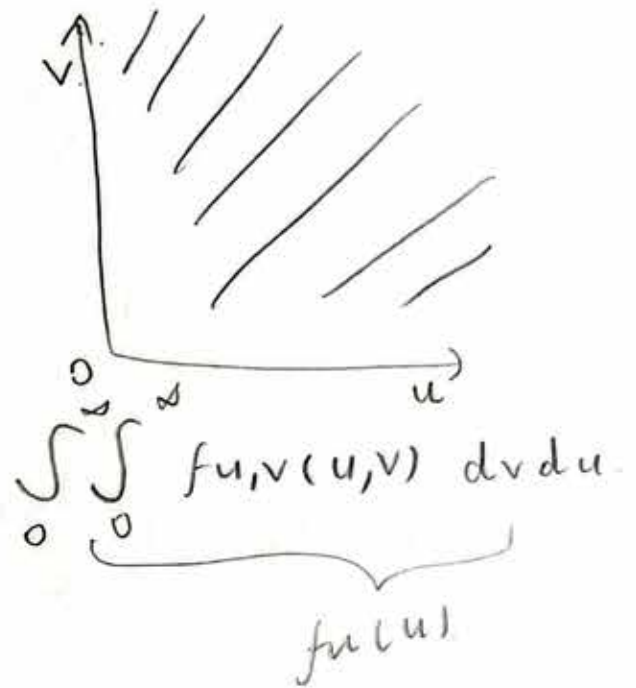
$$= \frac{1}{(u+1)^2} \left(-ve^{-v} - \int -e^{-v} dv \Big|_0^{\infty} \right)$$

$$= \frac{1}{(u+1)^2} \left(-ve^{-v} - e^{-v} \Big|_0^{\infty} \right)$$

$$= \frac{1}{(u+1)^2} (0 - (0 - 1))$$

$$= \frac{1}{(u+1)^2} \cdot 1$$

$$= \begin{cases} \frac{1}{(u+1)^2} & \text{si } u \geq 0 \\ 0 & \text{en o.c.} \end{cases} \quad \checkmark \text{ Integra a 1}$$



$$f_V(v)$$

$$= \int_0^{\infty} e^{-v} \cdot \frac{v}{(u+1)^2} du.$$

$$= v \cdot e^{-v} \int_0^{\infty} \frac{1}{(u+1)^2} du$$

$$\left| u = u+1 \right|$$

$$= v \cdot e^{-v} \int_0^{\infty} \frac{1}{u^2} du$$

$$= v \cdot e^{-v} \int_0^{\infty} u^{-2} du$$

$$= v \cdot e^{-v} \cdot \left(\frac{u^{-1}}{-1} \Big|_0^{\infty} \right)$$

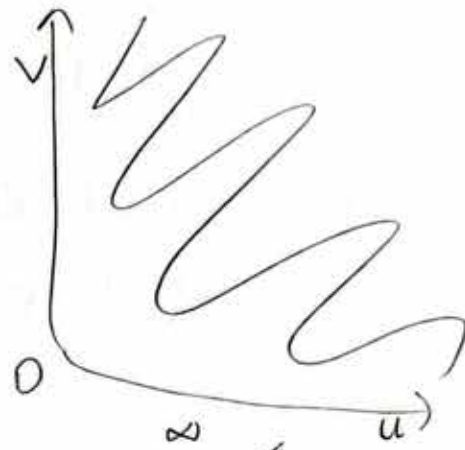
$$= v \cdot e^{-v} \left(-\frac{1}{u} \Big|_0^{\infty} \right)$$

$$= v e^{-v} \left(\frac{-1}{u+1} \Big|_0^{\infty} \right)$$

$$= v e^{-v} \left(0 + \frac{1}{1} \right)$$

$$= v e^{-v} \cdot 1$$

$$= \begin{cases} v e^{-v} & \text{si } v \geq 0 \\ 0 & \text{en o.c.} \end{cases}$$



$$\underbrace{\int_0^{\infty} \int_0^{\infty} f_{u,v}(u,v) du dv}_{f_V(v)}$$

✓ Integra a 1.

$$22. X \sim N(0, 1), Y \sim \Gamma\left(\frac{1}{2}, 2\right).$$

$$u = \frac{X}{\sqrt{Y}}, \quad v = Y. \quad \text{Determine } f_{u,v}(u, v) \text{ and } f_u(u).$$

$$f_X(x) = \begin{cases} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x^2}{2}} & -\infty < x < \infty \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} \cdot x^{\alpha-1} \cdot e^{-x/\beta} & y \geq 0 \\ 0 & \text{en o.c.} \end{cases}$$

$$\begin{aligned} \frac{\Gamma(1/2)}{\sqrt{\pi}} &= \frac{1}{\sqrt{2} \Gamma(1/2)} \cdot y^{-1/2} \cdot e^{-y/2} \\ &= \frac{1}{\sqrt{2\pi}} \cdot y^{-1/2} \cdot e^{-y/2} \end{aligned}$$

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{2\pi} \cdot e^{-\frac{x^2}{2}} \cdot y^{-1/2} \cdot e^{-y/2} & \begin{matrix} -\infty < x < \infty \\ y \geq 0 \end{matrix} \\ 0 & \text{en o.c.} \end{cases}$$

$$u = \frac{x}{\sqrt{y}} \quad v = x \quad \} \quad g^{-1}(u, v)$$

$$u\sqrt{y} = x.$$

$$(\sqrt{y})^2 = \left(\frac{x}{u}\right)^2$$

$$y = \frac{x^2}{u^2}$$

$$\frac{x^2}{u^2} = v.$$

$$\sqrt{x^2} = \sqrt{v \cdot u^2}$$

$$\pm |x| = \pm |u| \sqrt{v}.$$

$$x = u\sqrt{v}$$

$$h^{-1}(u, v).$$

$$|J| = \begin{vmatrix} \frac{dh^{-1}}{du} & \frac{dh^{-1}}{dv} \\ \frac{dg^{-1}}{du} & \frac{dg^{-1}}{dv} \end{vmatrix} \quad \begin{matrix} x = h^{-1} = u\sqrt{v} \\ y = g^{-1} = v \end{matrix} \quad \begin{matrix} \rightarrow u v^{\frac{1}{2}} \\ \frac{1}{2\sqrt{v}} \end{matrix}$$

$$= \begin{vmatrix} \sqrt{v} & \frac{u}{2\sqrt{v}} \\ 0 & 1 \end{vmatrix}$$

$$= \sqrt{v} - \frac{u}{2\sqrt{v}} \cdot 0 = \sqrt{v}$$

$$\boxed{|J| = \sqrt{v}}$$

$$x = h^{-1} = u\sqrt{v}$$

$$y = g^{-1} = v$$

$$f_{u,v}(u,v) = f_{x,y}(h^{-1}(u,v), g^{-1}(u,v)) \cdot |J|$$

$$= \frac{1}{2\pi} \cdot \frac{1}{\cancel{\sqrt{v}}} \cdot e^{\frac{(u\sqrt{v})^2}{2}} \cdot e^{-\frac{v}{2}} \cdot \cancel{\sqrt{v}}$$

$$= \int \frac{1}{2\pi} \cdot e^{-\frac{vu^2}{2}} \cdot e^{-\frac{v}{2}}$$

$$-\infty < u < \infty$$

$$v \geq 0.$$

0 en o. c.

✓ Integra
a 1.

$$-\infty < x < \infty$$

$$-\infty < u\sqrt{v} < \infty$$

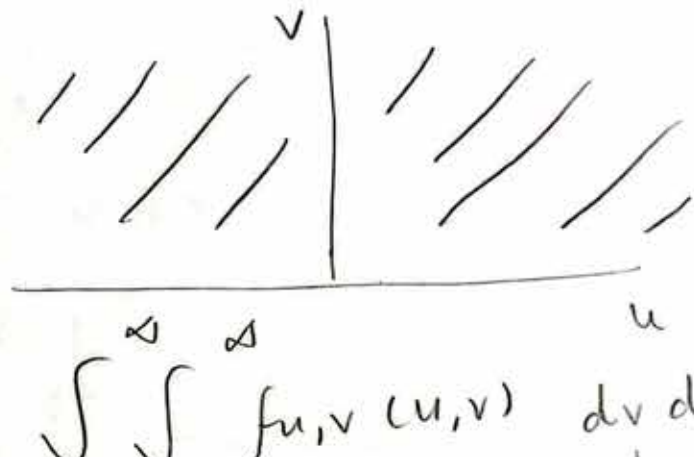
$$|-\infty < u < \infty|$$

$$v \geq 0.$$

$$|v \geq 0|$$

$f_u(u)$

$$= \int_0^{\infty} \frac{1}{2\pi} \cdot e^{-\left(\frac{1+u^2}{2}\right)v} dv$$



$$= \frac{1}{2\pi} \int_0^{\infty} e^{-\left(\frac{1+u^2}{2}\right)v} dv$$

$$\underbrace{\int_0^{\infty} \int_{-\infty}^{\infty} f_{u,v}(u,v) dv du}_{f_u(u)}$$

$$= \frac{1}{2\pi} \cdot \frac{e^{-\left(\frac{1+u^2}{2}\right)v}}{-\left(\frac{1+u^2}{2}\right)} \Big|_0^{\infty}$$

$$= \frac{1}{2\pi} \left(0 + \frac{1}{\frac{1+u^2}{2}} \right)$$

$$= \frac{1}{2\pi} \cdot \frac{2}{1+u^2}$$

$$= \begin{cases} \frac{1}{\pi(1+u^2)} & \text{if } -\infty < u < \infty \\ 0 & \text{en o. c.} \end{cases}$$

✓ Integra a 1.

23. $X, Y \sim N(0, 1)$.

$u = 2X - Y \quad v = -X + Y$.

Determine $f_{u,v}(u, v)$.

$$f_X(x) = \begin{cases} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x^2}{2}} & -\infty < x < \infty \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{y^2}{2}} & -\infty < y < \infty \end{cases}$$

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{2\pi} \cdot e^{-\frac{x^2}{2}} \cdot e^{-\frac{y^2}{2}} & \text{si } -\infty < x < \infty \\ & y - \infty < y < \infty \\ 0 & \text{en o. c.} \end{cases}$$

$$u = 2x - y \quad v = -x + y$$

$$u + y = 2x \quad v + x = \textcircled{y}$$

$$\textcircled{y} = 2x - u$$

$$2x - u = v + x$$

$$2x - x = u + v$$

$$x(2-1) = u + v$$

$$\textcircled{x} = u + v$$

$$h^{-1}(u, v)$$

$$u = 2x - y \quad v = -x + y$$

$$u + y = 2x \quad v + x = y$$

$$\frac{u+y}{2} = \textcircled{x}$$

$$\textcircled{x} = y - v$$

$$\frac{u+y}{2} = y - v$$

$$u + y = 2y - 2v$$

$$y - 2y = -2v - u$$

$$y(1-2) = -2v - u$$

$$-y = -2v - u$$

$$\textcircled{y} = 2v + u$$

$$u + y = 2y - 2v \quad g^{-1}(u, v)$$

$$u + 2v = 2y - y$$

$$u + 2v = y$$

$$|J| = \begin{vmatrix} \frac{dh^{-1}}{du} & \frac{dh^{-1}}{dv} \\ \frac{dg^{-1}}{du} & \frac{dg^{-1}}{dv} \end{vmatrix} \quad \left| \begin{array}{l} x = h^{-1} = u + v \\ y = g^{-1} = 2v + u \end{array} \right|$$

$$|J| = \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix}$$

$$|J| = 2 - 1 = 1$$

$$\boxed{|J| = 1}$$

$$f_{u,v}(u,v) = f_{x,y}(h^{-1}(u,v), g^{-1}(u,v)) \cdot |J|$$

$$= \frac{1}{2\pi} \cdot e^{-\frac{(u+v)^2}{2}} \cdot e^{-\frac{(2v+u)^2}{2}} \cdot 1$$

$$= \begin{cases} \frac{1}{2\pi} \cdot e^{-\frac{(u+v)^2}{2}} \cdot e^{-\frac{(2v+u)^2}{2}} & \text{si } -\infty < u < \infty, \\ & -\infty < v < \infty \\ & -\infty < x < \infty \\ & -\infty < u+v < \infty \\ & \boxed{-\infty < u < \infty} \\ & -\infty < v < \infty \\ & -\infty < 2v+u < \infty \\ & -\infty < 2v < \infty \\ & \boxed{-\infty < v < \infty} \end{cases}$$

✓ Integra a 1.

24. Si $Y \sim \Gamma(\alpha, 1)$, muestre que
 $W = \lambda Y \sim \Gamma(\alpha, \lambda)$.

Solucion

Por propiedad de escala.

Si $X \sim \Gamma(\alpha, \beta)$, para $c > 0 \Rightarrow cX \sim \Gamma(\alpha, c\beta)$.

Entonces:

Si $Y \sim \Gamma(\alpha, 1)$, para $\lambda > 0 \Rightarrow \lambda Y \sim \Gamma(\alpha, \lambda \cdot 1)$.
 $\therefore W = \lambda Y \sim \Gamma(\alpha, \lambda)$.

25. Si $Y \sim U(0,1)$, muestre que

$$W = -\ln(Y) \sim \text{Exp}(1).$$

Solucion

$$f_Y(Y) = \begin{cases} 1 & \text{si } 0 < Y < 1 \\ 0 & \text{en o.c.} \end{cases}$$

$$W = -\ln(Y).$$

$$\frac{dh^{-1}}{dw} = -e^{-w}$$

$$-W = \ln(Y)$$

$$\left| \frac{dh^{-1}}{dw} \right| = e^{-w}.$$

$$\underbrace{e^{-w} = Y}_{h^{-1}(w)}$$

$$h^{-1}(w).$$

$$f_W(w) = f_Y(h^{-1}(w)) \cdot \left| \frac{dh^{-1}}{dw} \right|$$

$$= e^{-w} \cdot 1$$

$$= \begin{cases} 1 \cdot e^{-w} & \text{si } w \geq 0 \\ 0 & \text{en o.c.} \end{cases}$$

$$\therefore W = -\ln(Y) \sim \text{Exp}(1)$$

~~$$W = -\ln(0)$$~~

~~$$W = -\ln(1)$$~~

$$W = 0$$

$$0 < Y < 1.$$

$$0 < e^{-w} < 1$$

~~$$\ln(0) < -w < \ln(1)$$~~

$$-w < \ln(1)$$

$$w > -\ln(1)$$

$$\boxed{w \geq 0}$$

26. si $Y \sim \text{Pareto}(\alpha, \beta)$

$$f_Y(y) = \frac{\alpha \beta^\alpha}{y^{\alpha+1}}, \quad y \geq \beta > 0, \alpha > 0$$

Muestre que $W = \ln\left(\frac{Y}{\beta}\right) \sim \text{Exp}(\alpha)$

Solución:

$$W = \ln\left(\frac{Y}{\beta}\right)$$

$$e^W = \frac{Y}{\beta}$$

$$\underbrace{\beta e^W = Y}_{h^{-1}(W)}$$

$$\begin{aligned} \frac{dh^{-1}}{dw} &= 0 \cdot e^w + \beta e^w \\ &= \beta e^w \end{aligned}$$

$$\left| \frac{dh^{-1}}{dw} \right| = \beta e^w$$

$$f_W(w) = f_Y(h^{-1}(w)) \cdot \left| \frac{dh^{-1}}{dw} \right| \quad \begin{aligned} Y &\geq \beta \\ \beta e^w &\geq \beta \\ e^w &\geq 1 \end{aligned}$$

$$= \frac{\alpha \beta^\alpha}{(\beta e^w)^{\alpha+1}} \cdot \beta e^w$$

$$= \frac{\alpha \beta^\alpha}{\beta^\alpha \cdot \beta^\alpha e^{w\alpha} \cdot e^w} \cdot \beta e^w$$

$$\begin{aligned} &= \frac{\alpha}{e^{w\alpha}} \quad \begin{aligned} w &\geq \ln(1) \\ W &\geq 0 \end{aligned} \\ &= \begin{cases} \alpha \cdot e^{-w\alpha} & \text{si } w \geq 0 \\ 0 & \text{en o. c.} \end{cases} \end{aligned}$$

$$\boxed{\therefore W \sim \text{Exp}\left(\frac{1}{\alpha}\right)}$$

27. Si $Y \sim \text{Exp}(\lambda)$. ¿Cómo se distribuye $W = 2\lambda Y$?

$$f_Y(y) = \begin{cases} \frac{1}{\lambda} \cdot e^{-y/\lambda} & \text{si } y \geq 0 \\ 0 & \text{en o.c.} \end{cases}$$

Solución

$$W = 2\lambda Y.$$

$$\underbrace{\frac{W}{2\lambda} = Y}_{h^{-1}(W)}.$$

$$\frac{dh^{-1}}{dw} = \frac{1 \cdot 2\lambda - \cancel{w \cdot 0}}{(2\lambda)^2}$$

$$= \frac{2\lambda}{(2\lambda)^2}$$

$$= \frac{1}{2\lambda}$$

$$\left| \frac{dh^{-1}}{dw} \right| = \frac{1}{2\lambda}$$

$$f_W(w) = f_Y(h^{-1}(w)) \cdot \left| \frac{dh^{-1}}{dw} \right|$$

$$= \frac{1}{\lambda} \cdot e^{-\frac{w}{2\lambda}} \cdot \frac{1}{2\lambda} \quad Y \geq 0$$

$$= \begin{cases} \frac{1}{2\lambda^2} \cdot e^{-\frac{w}{2\lambda^2}} & w \geq 0 \\ 0 & \text{en o.c.} \end{cases} \quad \begin{matrix} \frac{w}{2\lambda} \geq 0 \\ w \geq 0 \end{matrix}$$

$$\boxed{\therefore W \sim \text{Exp}(\frac{1}{2\lambda^2})}$$

Generatriz

27. Si $Y \sim \text{Exp}(\lambda)$.

¿cómo se distribuye $w = 2\lambda Y$?

Solución

Si $Y \sim \text{Exp}(\lambda)$

$$m_Y(t) = (1 - \lambda t)^{-1}$$

$$\text{Ahora, } m_w(t) = E(e^{tw})$$

$$= E(e^{t \cdot 2\lambda Y})$$

$$= E(e^{t' \cdot Y})$$

$$= m_Y(t')$$

$$= m_Y(2\lambda t)$$

$$= (1 - \lambda \cdot 2\lambda t)^{-1}$$

$$= (1 - 2\lambda^2 t)^{-1}$$

$$\boxed{\therefore w \sim \text{Exp}(2\lambda^2)}$$

si $X \sim \Gamma(\alpha, \beta)$.

$$m_X(t) = (1 - \beta t)^{-\alpha}$$

28. $X_1, \dots, X_n$. $X_i \sim \text{Exp}(\lambda)$.

¿cómo se distribuye $u = \frac{2}{\lambda} \sum_{i=1}^n X_i$?

Solución

¿cómo se distribuye $\sum_{i=1}^n X_i$?

$$m_{X_i}(t) = (1 - \lambda t)^{-1}$$

$$m_{X_n}(t) = (1 - \lambda t)^{-1} \cdot (1 - \lambda t)^{-1} \cdot (1 - \lambda t)^{-1} \dots$$

$$m_{X_n}(t) = (1 - \lambda t)^{-n}$$

$$\therefore X_i \sim \Gamma(n, \lambda)$$

Ahora, por propiedad de escala.

$$\text{si } X_i \sim \Gamma(n, \lambda)$$

$$\Rightarrow \frac{2}{\lambda} \cdot X_i \sim \Gamma(n, \frac{2}{\lambda} \cdot \lambda)$$

$$\Rightarrow u \sim \Gamma(n, 2)$$

$$\begin{aligned} m_u(t) &= (1 - 2t)^{-n} \\ &= (1 - 2t)^{-\frac{2n}{2}} \end{aligned}$$

$$\boxed{\begin{aligned} \therefore u &\sim \chi^2(2n) \\ \therefore u &\sim \Gamma(n, 2) \end{aligned}}$$

29. $X_1, \dots, X_n$.

$$f_X(x) = \frac{x}{\theta^2} e^{-\frac{x^2}{2\theta^2}}, \quad \theta > 0, x > 0.$$

¿cómo se distribuye $u = \frac{1}{\theta^2} \sum_{i=1}^n \underbrace{X_i^2}_V$?

Solución.

$$V = X^2$$

$$\sqrt{V} = \sqrt{X^2}$$

$$\sqrt{V} = |X|, \quad x > 0$$

$$\underbrace{\sqrt{V} = X}_{h^{-1}(V)}$$

$$\frac{dh^{-1}}{dv} = v^{\frac{1}{2}} = \frac{1}{2\sqrt{v}}$$

$$\left| \frac{dh^{-1}}{dv} \right| = \frac{1}{2\sqrt{v}}$$

$$f_V(v) = f_X(h^{-1}(v)) \cdot \left| \frac{dh^{-1}}{dv} \right|$$

$$= \frac{\sqrt{v}}{\theta^2} \cdot e^{-\frac{v}{2\theta^2}} \cdot \frac{1}{2\sqrt{v}}$$

$$= \begin{cases} \frac{1}{2\theta^2} \cdot e^{-\frac{v}{2\theta^2}} & \text{si } v > 0 \\ 0 & \text{en o. c.} \end{cases}$$

$$\begin{aligned} x &> 0. \\ (\sqrt{v})^2 &> (0)^2 \\ v &> 0^2 \\ \underline{v > 0} \end{aligned}$$

$$\therefore X^2 \sim \text{Exp}(2\theta^2)$$

$$\Rightarrow \text{si } X^2 \sim \text{Exp}(2\theta^2).$$

$$\Rightarrow m_{X^2}(t) = (1 - 2\theta^2 t)^{-1}$$

$$\Rightarrow m_{X^2_n}(t) = (1 - 2\theta^2 t)^{-1} \cdot (1 - 2\theta^2 t)^{-1} \dots$$

$$= (1 - 2\theta^2 t)^{-n}$$

$$\therefore \sum_{i=1}^n X_i^2 \sim \Gamma(n, 2\theta^2)$$

Ahora, por propiedad de escala:

$$n \sum_{i=1}^n X_i^2 \sim \Gamma(n, 2\theta^2)$$

$$\Rightarrow \frac{1}{\theta^2} \sum_{i=1}^n X_i^2 \sim \Gamma(n, \frac{1}{\theta^2} \cdot 2\theta^2)$$

$$\Rightarrow u \sim \Gamma(n, 2)$$

$$\therefore m_u(t) = (1 - 2t)^{-n}$$

$$= (1 - 2t)^{-\frac{2n}{2}}$$

$$\therefore u \sim \chi^2(2n)$$

30. $X_1, \dots, X_n \sim \text{Pareto}(\alpha, \beta)$.

$$f_Y(y) = \frac{\alpha \beta^\alpha}{y^{\alpha+1}}, \quad y \geq \beta > 0, \alpha > 0$$

a) Muestre que el mínimo muestral $Y_{(1)} \sim \text{Pareto}(n\alpha, \beta)$.

$$\begin{aligned} F_Y(y) &= \int_{\beta}^y \frac{\alpha \beta^\alpha}{t^{\alpha+1}} dt \\ &= \alpha \beta^\alpha \int_{\beta}^y \frac{1}{t^{\alpha+1}} dt \\ &= \alpha \beta^\alpha \int_{\beta}^y t^{-\alpha-1} dt \\ &= \alpha \beta^\alpha \left. \frac{t^{-\alpha}}{-\alpha-1+1} \right|_{\beta}^y \\ &= \alpha \beta^\alpha \left. -\frac{1}{\alpha t^{\alpha}} \right|_{\beta}^y \\ &= \alpha \beta^\alpha \left(-\frac{1}{\alpha y^{\alpha}} + \frac{1}{\alpha \beta^{\alpha}} \right) \\ &= -\frac{\alpha \beta^{\alpha}}{\alpha y^{\alpha}} + 1 \quad \left| = -\frac{\beta^{\alpha}}{y^{\alpha}} + 1 \right| \end{aligned}$$

$$f_{Y_{(1)}}(y) = n(1 - F_Y(y))^{n-1} \cdot f_Y(y)$$

$$= n \left(1 - \left(\frac{\beta}{y} \right)^\alpha \right)^{n-1} \cdot \frac{\alpha \beta^\alpha}{y^{\alpha+1}}$$

$$= n \left(\left(\frac{\beta}{y} \right)^\alpha \right)^{n-1} \cdot \frac{\alpha \beta^\alpha}{y^{\alpha+1}}$$

$$= n \left(\frac{\beta}{y} \right)^{n\alpha - \alpha} \cdot \frac{\alpha \beta^\alpha}{y^{\alpha+1}}$$

$$= \frac{\alpha n \cdot \beta^{n\alpha} \cdot \cancel{\beta^{-\alpha}} \cdot \cancel{\beta^{-\alpha}}}{y^{n\alpha} \cdot \cancel{y^{-\alpha}} \cdot \cancel{y^{-\alpha}} \cdot y^1}$$

$$= \frac{\alpha n \cdot \beta^{n\alpha}}{y^{n\alpha+1}}$$

$$\therefore Y_{(1)} \sim \text{Parceto}(\alpha n, \beta)$$

b) ¿cómo se distribuye $u = \frac{Y_{(1)}}{\beta}$?

Solución

$$u = \frac{Y_{(1)}}{\beta}$$

$$\underbrace{u\beta = Y_{(1)}}_{h^{-1}(u)}$$

$$\frac{dh^{-1}}{du} = \beta$$

$$\left| \frac{dh^{-1}}{du} \right| = \beta$$

$$f_u(u) = f_Y(h^{-1}(u)) \cdot \left| \frac{dh^{-1}}{du} \right|$$

$$= \frac{\alpha n \cdot \beta^{\alpha n}}{(u\beta)^{\alpha n+1}} \cdot \beta$$

$$= \frac{\alpha n \cdot \beta^{\alpha n}}{u^{\alpha n} \cdot u' \cdot \beta^{\alpha n} \cdot \beta'} \cdot \beta$$

$$= \frac{\alpha n}{u^{\alpha n+1}}$$

$$= \frac{\alpha n (1)^{\alpha n}}{u^{\alpha n+1}}$$

$$\therefore U \sim \text{Pareto}(\alpha n, 1)$$